International Labour Organization

Reference guide on Employment Impact Assessment

Second edition

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Introduction on employment impact assessments

Reference guide on employment impact assessment

Main points

- The International Labour Organization uses employment impact assessments to support evidence-based policy advice that enhances job creation, employment conditions and workers' employability.
- The ILO takes a flexible approach on employment impact assessments, relying on diverse methodologies, ex ante and ex post, and tailored to specific policy needs as well as resource and time constraints.
- The ILO approach highlights the need to focus on the quantity and the quality of jobs; to assess if interventions are pro-employment; and to advocate for social dialogue among social partners in this process.
- Institutionalizing employment impact assessments within national frameworks and budgetary processes is key to ensuring that assessments lead to informed policy decisions and sustainable employment outcomes.

The reference guide[1]

The main work of the Employment Policy, Job Creation and Livelihoods Department (known as EMPLOYMENT) of the International Labour Organization involves advice to constituents on employment policies at the macro, meso and micro levels. This policy advice is guided by the ILO principles enshrined in its Conventions and Recommendations. Such advice must be agile and responsive to different scenarios, thus requiring ready-to-be-applied methods and tools for assessing impacts and thus improves the understanding of what works, as well as when, where and how it works.

1 Members of the Task Force on Employment Impact Assessments of the ILO Employment Policy, Job Creation and Livelihoods Department authored this reference guide: Sangheon Lee, Juan Chacaltana, Matthieu Charpe, Sara Elder, Christoph Ernst, Drew Gardiner, Jean Francois Klein, Ralf Krueger, David Kucera, Massimiliano La Marca, Niall O'Higgins, Dorothea Schmidt-Klau, Bolormaa Tumurduchur-Klok and Xiao Jiang. The 13-part reference guide also benefited from the excellent contributions of Alina Game, Luis Villanueva, Gabriel Michelena, and Jürgen Weller.

Due to the complex nature of employment policies, including macro, industrial and specific interventions implemented in different scenarios, the EMPLOYMENT Department uses a variety of impact assessment approaches for its evidence-based policy development: qualitative, quantitative, *ex ante*, *ex post*, macro, sectoral and individual. All these dimensions are relevant for assessing the success of an intervention.

Without solid impact assessments, the credibility and reliability of policy orientation efforts can be challenged, with significant implications for the validity and accountability of the ILO's work, as well as for resource mobilization. The 13-part reference guide covers guidelines for the EMPLOYMENT Department on employment impact assessments. This first part (section) introduces the concept of an employment impact assessment and the 11 methods or tools most frequently used by the EMPLOYMENT Department, each of which is further articulated in a stand-alone document. A twelfth section is included to discuss impact assessments of informal employment. These individual guides are the result of discussions within the Task Force on Employment Impact Assessments, established within the EMPLOYMENT Department to review the practice and tools and provide recommendations for strengthening what is being done currently.

Each part describes a specific method and discusses the frequently asked questions that specialists in the field and at headquarters receive. This is an updated version of the employment impact assessment methodologies at EMPLOYMENT, with three new sections. The purpose of this reference guide is to elaborate how the Department uses or should use these methodologies.

The ILO – through its EMPLOYMENT and other departments – is not the only institution producing evidence on employment impacts. It is a highly competitive area, with universities, non-government organizations and other international institutions also producing an increasing amount of information on the subject. The Department's leading role in employment impact analysis is critical for ensuring continuity of the ILO's global leadership in employment policies.

▶ Concept and approaches of an impact assessment

Concept

The rationale for an employment impact assessment is to anticipate and/or measure the potential or actual impact of policies or interventions on jobs and job-related outcomes as an ongoing process to constantly improve the design and implementation of policy and programme options. Employment policy is an area wherein there are relatively large gaps in evidence. An employment impact assessment can contribute towards improving the effectiveness of policy responses. Linked to this is the complexity of the policymaking process, especially when related to employment and labour market outcomes, where stakeholders may have diverging interests. This is where the ILO has a particularly important role in building up the institutional capacity of national employment authorities and other institutions to conduct these types of assessments and to advise on institutional coordination mechanisms.

Approaches

The ILO provides strategic advice to governments, businesses and workers on the development and implementation of effective employment policies. This guidance is essential for enabling policymakers to make informed decisions that foster job creation and enhance working conditions. The ILO's approach on employment impact assessments is characterized by its flexibility, relying on a diverse array of quantitative and qualitative tools. These instruments are designed to assess intervention impacts at

various stages, whether before (*ex ante*) or after (*ex post*) implementation, and to operate across different levels of intervention – macro (national), sectoral (industry-specific) and micro (individual businesses).

The EMPLOYMENT Department uses a series of methods and tools for assessing employment impact. A consultation with colleagues in headquarters and field employment specialists on their use of these methods led to a listing of more than 100 research study documents directly or indirectly developed by the Department (see the annex) that reflect use of an employment impact assessment. Analysis of these documents revealed the following approaches as the most frequently used.

General equilibrium model

- A general equilibrium model is typically used to assess the economy-wide effects of policy changes by accounting for the interdependence among all sectors, markets, regions or countries and different groups of workers or households. There are several types of this model, including the computable general equilibrium model and the dynamic stochastic general equilibrium model.

Input–output table

- This tool is based on the information derived from input–output tables, which provide a comprehensive overview of all inter-industrial sectoral flows within a national economy. This tool is typically used ex ante to assess the impact of a proposed policy on employment, output and value added across economic activities. The tool typically produces static outcomes in terms of direct, indirect and induced employment effect. Nevertheless, it is feasible to utilize a dynamic, albeit less prevalent, version of the model. Input–output analysis has been widely used at the ILO to assess the quantity of employment generated as well as the type and quality of employment (for example, informal employment or green jobs) by opening the labour demand in different categories, such as gender, age, skill level and formal status.

Social accounting matrix

- A social accounting matrix represents an extension of the input–output table, combining intersectoral flows with data from national accounts, the balance of payments and household surveys. This enables the incorporation of a comprehensive matrix of monetary flows in an economy, encompassing transactions across all relevant institutional sectors, including the government, enterprises, the rest of the world and households. In this approach, there are no black holes – every purchase necessarily implies a sale, and every payment necessarily implies an income (Lavoie and Godley 2007). The ILO has used this methodology extensively in recent decades to measure, among other things, the employment effects of public investment through multiplier analysis.

Growth decomposition

- Several empirical studies have used the growth accounting decomposition method to evaluate the relative importance of employment reallocation and within-sector effects, as well as the sectoral drivers of aggregate productivity growth, particularly labour productivity growth. These methods can reveal patterns that are not otherwise readily apparent, thereby helping to increase the understanding of processes of structural transformation. They can also usefully inform sectoral policies by clarifying whether and how sectoral drivers within more- and less-successful countries differ from each other.

Local multiplier

- The local multiplier method measures the employment potential of different economic sectors, relying on changes in employment across two census data sets. This method has a dynamic dimension and a geographical dimension, enabling the construction of a cross-section or panel data of sectoral employment (tradable, non-tradable) across administrative entities. The method looks directly at employment (whereas input–output analysis goes from production multipliers to employment multipliers). Because it is based on census data, the local multiplier methodology additionally allows for a thorough statistical analysis even in small areas.

Sector identification

- Renewed interest in a new generation of industrial policies is motivated by the belief that well-designed and implemented industrial policies are essential to sustainable structural transition processes. What methodologies exist to select the "right" industries? Three methods help to determine potentially promising products and corresponding sectors for export expansion and structural transformation: (i) the growth identification and facilitation framework; (ii) the economic complexity and product space method; and (iii) the International Trade Centre's export potential and product diversification indicators.

Structural model for sustainable development

- This is a quantitative simulation tool for assessing the sector-disaggregated, economy-wide impacts of interventions, such as policies or specific investments. The model is based on a structuralist approach. Such a method focuses on the structure of the economy, the role of its institutions, the socio-economic actors and their economic relationships. In this model, sectoral details can include effects on different products, activities, labour types (age group, gender, relative wages or earnings, employment and formality status, skills and occupations), households (by income levels and rural versus urban) and other institutions, hence allowing for different types of economic analysis and their social and environmental extensions.

Skills anticipation

- Skills anticipation requires converting information on job projections into skills needs. Other pieces of information are needed, including employment by sector and occupation. Some studies focus on qualitative data (from focus groups, round tables, expert interviews, foresights and scenario development); surveys among employers (establishments or enterprises); or tracer studies of school or training graduates and school-to-work transition surveys.

Experiment (randomized controlled trial)

- An experiment or randomized controlled trial assesses the effectiveness (did it work or not?) and causality (does it work because of the intervention?) at the same time. The usual approach is to compare participants with non-participants to see differences. Because these differences can be attributed to the impact of the intervention or to pre-existing differences (selection bias), this method uses a random process to decide who is granted access to the programme or intervention and who is not, so that every eligible unit has the same probability of being selected for treatment. Together with partner institutions, the EMPLOYMENT Department has implemented impact evaluations using randomized controlled trials, including through the Taqeem Initiative.

Quasi-experiment

- There are other ways to control selection bias that are valid under certain assumptions (Duflo 2006). Several techniques can be used for this, including matching methods, difference-in-difference, instrumental variables and regression discontinuity design (World Bank 2016). A recent example from the EMPLOYMENT Department is the application of a double- and triple-difference estimation of the impact of the youth guarantee scheme in Italy (O'Higgins and Pica 2017). Other applications have been implemented by, for example, the Taqeem Initiative (Elsayed and Roushdy 2017).

Meta-analysis

- A systematic review summarizes the results of available and carefully crafted studies (controlled trials) and provides a high level of evidence regarding the effectiveness of interventions. Judgements may be carried out on the evidence and to inform recommendations for employment and other relevant fields. These reviews are complicated and depend largely on what trials are available, how they were implemented (the quality of the trials) and the outcomes that were measured. Review authors pool numerical data about the effects of the intervention via a process called "meta-analysis". Subsequently,

they examine that evidence for any benefit and/or harm from those treatments. Thus, systematic reviews summarize the existing research on a topic.

Geographic information system and remote sensing

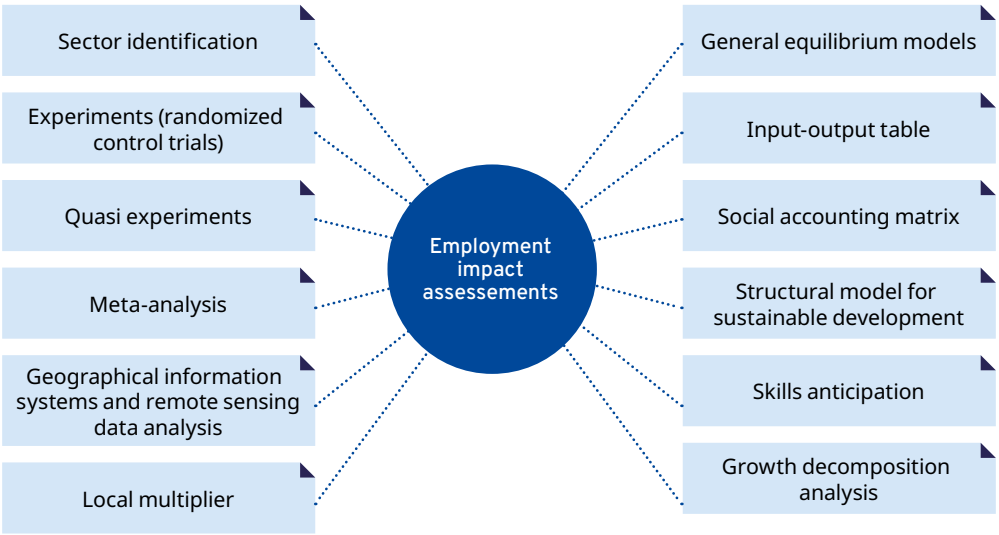
- Geographic information system and remote sensing methods harness such information as night-light data to monitor economic activities in data-scarce regions. By combining geographic data with survey information, these methods provide evidence to assess employment outcomes and policy interventions, including for discussions about causality. For instance, remote sensing data were used to evaluate road infrastructure projects in Kenya and Rwanda, the impact of drought in Zambia on agricultural productivity, the impact of conflicts in Myanmar on the labour market and the level of inequalities in low-income countries and dynamics of productivity in sub-Saharan countries.

Informal employment impact assessment

- Included as a separate section in this series is an explanation of an informal employment impact assessment, which parallels traditional employment impact evaluation, in many of the methods and tools used. However, as the guide points out, it is crucial to also consider the qualitative dimensions of employment because they offer nuanced insights into the heterogeneity and complexity inherent in informal labour markets, thereby enriching the overall analytical framework.

In practice, all these approaches have strengths and weaknesses. Some are better equipped for *ex ante* analysis and others for *ex post* analysis. Some are better for revealing the impact on beneficiaries – workers or businesses – and others for reporting the impact on the economy or economic environment. The following diagram illustrates the main application of each method or tool – whether primarily applied *ex ante* or *ex post*.

A simple typology of the ILO's more common employment impact assessment approaches



Source: Authors.

This typology is a simplification. The image here reflects what is most commonly used in the ILO. In many cases, applications combine different methods or tools, for example, quantitative and qualitative methods, which are complementary and together can provide more in-depth insights into employment effects.[2]

▶ ILO approach on employment impact assessments

The ILO's practices with employment impact assessments are shaped by the types of interventions that are assessed: polices, investments, programmes and initiatives whose objective is to produce a change in employment outcomes, such as increasing job opportunities, improving employment conditions, enhancing workers' employability or mitigating negative impacts on employment caused by economic disruptions or external shocks.

In the ILO practice – reflected extensively in the EMPLOYMENT Department studies listed in the annex – the most common interventions whose impacts have been assessed are related to:

- Infrastructure projects
- Active labour market policies and programmes, including skills and employment services
- Demographic targeted policies, such as youth or gender employment policies
- Sectoral policies
- Macroeconomic policies
- Legislative proposals or reforms

These types of employment-related interventions – which many studies demonstrate – can be significantly influenced by the characteristics of the participants as well as a specific sector, region or industry the intervention targets and even by broader economic conditions or labour market dynamics. The specialized literature labels this as «heterogeneity» in employment effects, which means that the impact of an employment intervention may vary depending on the characteristics of the participants or the context in which it is implemented. Even if some methodologies allow ruling out the influence of multiple factors that might otherwise affect outcomes to thus isolate the causal impact of the intervention, in practice, knowing how the context (for example, the specific sector or region in which a programme is implemented) influences a programme's effectiveness can help policymakers design interventions that are more likely to succeed in diverse settings. Thus, a nuanced approach is often recommended, including the use of different and complementary methodologies.

Another important element of the ILO practices relates to the indicators used. The more common indicators are:

- Quantity of jobs, which includes the number of persons employed or, in some cases, the number of annual equivalent jobs
- Quality of jobs, which includes indicators for working poor, vulnerable employment or informal employment

2 Shaffer (2018) discussed some of the pros and cons of mixed methods, in particular regarding the precision of qualitative studies, external validity and usefulness in terms of causal inference. This reference guide also discusses the pros and cons of each method.

- Labour intensity
- Duration of employment
- Unemployment
- Employability
- Productivity

Gender breakdowns are essential. In practice, there has been emphasis on the quantity of jobs as a primary indicator, especially the number of employed persons. This indicator is comparatively easy to quantify, so virtually all approaches reviewed here can measure it. However, within the ILO (1999), there is concern "with decent work, and the goal is not just the creation of jobs but the creation of jobs of acceptable quality. The quantity of employment cannot be divorced from its quality" (ILO 1999).[3]

Thus, additional indicators should be used to complement the quantity dimension, especially indicators related to income, quality or composition of employment.[4] Options include:

- Income, which is also quantifiable, and methods are capable of measuring income effects.
- Indicators of quality of employment (working poor, vulnerable employment or informality), although this is more complex and requires more data, especially in macro or input–output measurements.
- The composition of employment by occupation, educational attainment or skill level.

Due to the nature of employment interventions and specific requests the ILO receives, there is no one method or tool that fits all scenarios. Some are more fit to respond to specific questions than others. For example, investment projects (such as for infrastructure) tend to be assessed with input–output multipliers, local multipliers or geographic information system approaches. Macro or sectoral policies are more related to general equilibrium or partial equilibrium models. Interventions more closely related to active labour market programmes, skills and labour market or other reforms have been particularly assessed with experiments, quasi-experiments or meta studies.

The ILO's EMPLOYMENT Department relies on the diverse array of methods and tools, which have mostly been developed by academics. The Department opts for several features in its employment impact assessments exercises:

i. The Department seeks (or should) to ensure that the theoretical background of any employment impact assessment is pro-employment, in the sense that employment is not considered a residual but a central target of the intervention.

ii. The Department seeks (or should) to inform and ensure the participation of social partners in all steps of an impact assessment process, from the design to the results stage, to promote social dialogue.

iii. The Department targets the quantitative employment impacts but should also provide some insight on the quality of that employment.

3 Decent work sums up the aspirations of people in their working lives. It involves opportunities for work that is productive and delivers a fair income, security in the workplace and social protection for families, better prospects for personal development and social integration, freedom for people to express their concerns, organize and participate in the decisions that affect their lives and equality of opportunity and treatment for all women and men (ILO 1999).

4 It is difficult to assess quality only when based on one indicator, but the idea is not to use only quantity-related indicators.

▶ ILO in action: Recent technical cooperation work on employment impact assessments

Over the past few decades, as previously explained, the ILO has produced more than a hundred studies using an employment impact assessment (see the annex) and has engaged in numerous technical cooperation projects aimed at employment-related challenges.

One notable example is the Taqeem Initiative, implemented by the ILO in collaboration with various partners to enhance youth employment in the Middle East and North Africa through rigorous monitoring and evaluation methodologies. Spanning two phases, from 2010 to 2018, the project helped develop evidence-based policies by equipping youth employment organizations with the tools and knowledge needed to assess the impact of their programmes. During the first phase (2010–13), Taqeem partnered with 15 youth employment organizations, reaching approximately 15,000 young people annually. The project facilitated "evaluation clinics" that provided training on impact evaluation techniques and introduced technology-enhanced, cost-effective monitoring tools. These efforts were complemented by the creation of a Web-based knowledge-sharing platform and the provision of small grants to support evaluation activities. Randomized controlled trials were conducted to validate the effectiveness of various programmes, contributing to a more robust evidence base.

The second phase (2013–18) focused on integrating the evaluation clinic model into local research institutions to establish "monitoring and evaluation hubs" for youth-serving organizations. This phase emphasized generating evidence on effective interventions through impact evaluations and disseminating findings to policymakers, institutions and investors to inform decision-making processes. The project achieved significant milestones, including the development of an impact evaluation system for youth-led organizations, the pioneering of studies on youth apprenticeship schemes, the economic empowerment of adolescent girls and the training of more than 500 youth practitioners. The project promoted the use of information and communication technology for data collection, transitioning from paper-based to digital tools. The dissemination of evaluation results through the Taqeem portal and social media enhanced the visibility and impact of the project's findings, ultimately influencing policy and investment decisions and leading to a national scale-up of successful programmes.

Other important technical cooperation projects include STRENGTHEN2 and METI, which succeeded the original STRENGTHEN Project in terms of employment impact assessments. METI included employment impact assessments as one outcome, but STRENGTHEN2 was entirely dedicated to assessing employment impact. Combined, these projects contributed to advancements in various methodologies, including those based on geospatial information system methods, and undertook numerous assessments at the project and portfolio levels.

Methodological advancements achieved through these projects include:

- Development of three geographic information system-based approaches.
- Application of structuralist computable general equilibrium models to investment projects.
- Advances in data for simpler input–output multiplier analysis.

These methodological advancements were applied to a larger number of practical employment impact assessments than ever before, focusing primarily on the transport and energy sectors. These assessments were mostly in-depth at the project level, with a few portfolio-level assessments creating input–output-based estimates for many projects to highlight common features and differences across project types and geographies.

Engagements with international finance institutions were conducted at a technical level to advance discussions on measuring employment impacts (with insights from how they measured their impacts

of their operations) and to give these impacts more consideration throughout the life cycle of the operations. Discussion points included:

- New methods to quantify employment impacts.
- Further harmonization and refinement of existing methods and tools (such as input–output-based multipliers).
- Advancements in discussions on indicators to measure the quality of employment generated by the international finance institutions' operations.

The most comprehensive collaboration took place with the African Development Bank to establish a Youth, Jobs and Skills Marker System. This system emphasizes youths, jobs and skills considerations in all of the Bank's operations, particularly during the planning, inception, design and appraisal stages of the project life cycle.

Institutionalization of employment impact assessments

The institutional capacities and processes required to carry out and use the results of an employment impact assessment is an important challenge. Without sufficient capacity and processes, these assessments may not be feasible in the first place or – even if an assessment is successfully concluded – may not result in further action.

Due to the complex nature of methodologies, sometimes agencies cannot carry out an impact assessment and thus rely on specialized institutions or hired specialists. Some level of internal technical capacity is desirable, in the medium term, at the least. The following describes some alternative possibilities for this.

An employment impact assessment can be included as a task of a more general social monitoring and evaluation institution. In Portugal, for instance, the Central Bank, together with partner institutions, evaluates the quantitative impact of structural reforms on employment and other outcomes. It uses econometric approaches and a dynamic stochastic general equilibrium model to assess the impact of the National Reforms Programme on employment, productivity and inclusiveness (Banco de Portugal, 2021).

In other cases, there are specialized entities because specific expertise is required to conduct a particular assessment.

- The Republic of Korea installed the Center for Employment Impact Assessment at the Korean Labor Institute as the designated body in charge of assessments. As of July 2019, a total of 819 programmes had been assessed, including 180 labour market policy programmes, 349 research and development programmes, 30 programmes in the arts, culture and sports sectors and 42 welfare and public health programmes (Lee 2019; Yoon 2019).
- The United Kingdom's Department for Work and Pensions is a leader in evaluating national active and passive labour market policies. Most evaluation activities are delivered by external contractors, and the Department commits a high level of funding and staff time to supporting evaluations. A range of methodologies are employed in the Department's evaluations, ranging from qualitative interviews to the use of more sophisticated quasi-experiment methods and randomized controlled trials to identify gross and net impacts. All evaluations are made available to the public, including those with negative results (Department of Work and Pensions, 2021).

- In Germany, the Institute for Employment Research conducts research on labour market issues to advise political actors at all levels. Economists, sociologists and researchers from other disciplines create the base for empirically informed labour market policy (European Commission, 2022). The Institute was founded in 1967 and has been a special office of the Federal Employment Agency since 2004. With the institutionalization as a special office, a clear separation was made between the Institute and the Federal Employment Agency, although the Institute was firmly anchored within the Executive Board of the Federal Agency. This organizational structure emphasizes the Institute's significance as the central institution for science-based policy advice on issues related to the labour market (Institut für Arbeitsmarkt- Und Berufsforschung, 2024).

Employment impact assessments have been developed in a priority sector supported by a dynamic agency of a government, sometimes with strong donor support and important policy questions to be addressed. Robalino and Lieuw-Kie-Song (forthcoming) featured examples in this direction: In Peru, for example, the Ministry of Finance leads the move towards results-based budgeting, with the Ministry of Labour deploying assessments to evaluate employment impacts. In Rwanda, the Ministry of Public Service and Labour leads the employment impact assessments efforts, supported by the Ministry of Finance in an initiative that has become more cohesive since the 2019 National Employment Policy, aiming for a pro-employment budget. In Cameroon, employment impact assessments are used to monitor the impacts of the National Development Strategy, with the Ministry of Economy, Planning and Land Planning taking the lead on the monitoring of that strategy.

There is also the possibility of an ad hoc approach. This usually evolves from individual evaluations carried out to take advantage of available funding or originating from the desire of a government official or a specific donor.

To enhance effectiveness, it is recommended to gradually adopt and institutionalize these assessments within national policy frameworks and budgeting processes. Ensuring integration with national employment policies and results-based budgeting will align public expenditures with employment outcomes more effectively. Additionally, leveraging employment impact assessments to inform both *ex ante* and *ex post* evaluations and using simulation models to project and set employment targets will improve policy decisions and budget allocations, ultimately fostering better labour market outcomes (Robalino and Lieuw-Kie-Song, forthcoming).

The ILO supports its constituents, with emphasis on governments, to leverage employment impact assessments as a means for generating insights in support of evidence-based policy and investment decision-making. Ultimately, the goal is for countries to undertake such assessments on their own and use the results to inform their decision-making. This requires awareness and support from the top, an institutional set-up that facilitates consensus politically and ensures a minimum level of technical capacity to generate the necessary data and higher education curricula that include related content – all of which will ensure the sustainability of a national employment impact assessment ecosystem.

▶ What has been learned?

In practice, successful implementation of employment impact assessments depends on a range of factors, three of which are highlighted here.

Budget and time constraints

The budget and time available for employment impact assessments is likely to be an important determinant of the applied approach. For example, a qualitative assessment is an important low-resource-intensive method. Some partial equilibrium models, local multipliers or sectoral analysis require moderate resources. In general, most approaches are expensive to develop in the first place and/or tailor to the needs of individual policies, if major changes to the model and available data are required (once set up, the marginal cost of running models can be modest). Experiments and quasi-experiments are also considered resource-intensive. From the policy perspective, a critical question is how the effects of a particular policy initiative will be felt over time (in the short, medium and longer terms). Some methods and models can only provide information on a newly emerging equilibrium at a specific point in time after the policy has been implemented. Other methods and models can take into account the reaction of affected populations to the policy change and thus provide an indication of the long-term effects of an intervention in question. It is therefore imperative to choose when to evaluate.

The link between monitoring and evaluation

Does every intervention need to be assessed in terms of impact? Not necessarily. The monitoring stage can provide important information for the development of an effective employment impact assessment. Monitoring assesses how an intervention evolves over time, while an evaluation looks at data in terms of initial goals, indicators and outcomes associated with a programme. Although monitoring does not address the issue of causality, it helps to understand if the intervention was implemented according to its theory of change, or the causal logic of how and why a particular programme, programme modality or design innovation will reach its intended outcomes. Take the case of the processes needed to produce results in a typical training programme: For a training to produce results, it first must take place (one party transferring messages to others). Then those messages must be incorporated by their receivers (learners). And then that information or knowledge must change some behaviour among those learners. Only if that happens will the capacity of an individual to be employed (employability) have been transformed. But even if this happens, it does not guarantee that employment will be created because it does not only depend on training.

Another case concerns infrastructure projects. Here, the exogenous variable is investment, which is typically public. This investment first must be implemented, which involves the construction of the infrastructure. That process will generate both direct employment (in the construction sector) and indirect employment (in suppliers, for example). This short-term effect on employment can be considered an output of a project. However, sustained investment can produce long-term impacts (increased employment in the construction sector). In addition, the infrastructure itself, if relevant, can produce changes in the productive structure of a community or locality that could boost permanent employment creation. Solid monitoring systems can resolve the debate on attribution versus contribution. Contribution is related to the processes or outcomes in control of the intervention. But for impact to happen, other elements need to be in place. On the other hand, if monitoring reports do not indicate that intermediate processes took place, then it is almost certain that no impacts were realized.

Learning from employment impact assessments

The value of employment impact assessments is a subject of increasing interest, especially by policymakers and practitioners. In theory, such an assessment should provide crucial information for improving interventions related to employment outcomes. However, some empirical studies have found that in practice this might not be happening. For example, in some cases, impact assessment exercises can be driven by donor requirements rather than a genuine desire for evidence-based policy.[5] To bridge this gap, some literature suggests a shift towards fostering a more strategic and evidence-based approach to using employment impact assessments instead of a merely checkbox exercise to create a learning loop to refine and improve employment interventions. This requires further collaboration and capacity-building among policymakers, researchers, implementing agencies and social partners.

▶ Going forward

The effectiveness of employment programmes or interventions is highly contingent upon contextual factors. Even approaches directly addressing issues of attribution and causality find recurrent evidence on heterogeneity. Consequently, understanding the full efficacy of a programme requires both determining whether it works or not and elucidating the mechanisms by which it operates, including its impacts across diverse contexts and the potential for adaptation. Hence, a nuanced and adaptable approach, employing a spectrum of methodologies, is advocated to generate robust and comprehensive evidence to inform policy decisions.

In practice, the application of employment impact assessment methods and tools face several challenges that deserve further attention. Notable areas for improvement include:

- Integration of learning from empirical evidence, such as going beyond checkbox approaches to incorporating insights gained from previous empirical studies into future policy assessments.
- Institutionalization, such as establishing robust frameworks adapted to various contextual settings, which seems to work better when employment impact assessments are connected to budgeting processes.

In general, improving employment assessment sustainability requires genuine evidence uptake in employment policy formulation. This, in turn, requires addressing both the supply and demand sides of the evidence system. On the supply side, producers of evidence (research institutions, think tanks, policy-oriented organizations, etc.) must prioritize generating high-quality, rigorous evidence that is relevant to policymakers' needs. This includes strengthening research capacity, fostering partnerships between local and international institutions and enhancing the ability of local actors to conduct impact evaluations and policy-relevant research. On the demand side, it is crucial to build up the capacity of policymakers and public officials to access, interpret and apply research findings in their decision-making processes. This can be achieved through targeted capacity-building initiatives, such as training on evidence-based policymaking, and by embedding research intermediaries within government agencies to facilitate ongoing dialogue between researchers and policymakers. Additionally, creating structured platforms for knowledge exchange, including policy dialogues, working papers and policy briefs, can improve the communication of research findings. Aligning research agendas with policy priorities and ensuring timely and context-sensitive dissemination of evidence will further enhance the integration of evidence into employment policy development.

5 See Manning et al (2020).

There has been an evolution of employment impact assessment methodologies. Initial approaches, such as the input–output analysis, concentrated on analysing inter-industry dependencies to estimate employment changes resulting from policy shifts. Subsequently, the social accounting matrix extended this approach by incorporating income distribution and social aspects for more holistic analysis. More recently, the randomized controlled trials and quasi-experiment designs have introduced rigorous causal inference into assessments. Emerging technologies, such as dig data analytics, machine learning, spatial data analysis, remote sensing and satellite imagery, are now significantly transforming employment impact assessments. The integration of these innovative technologies with traditional methodologies can facilitate a more nuanced and data-driven understanding of the intricate relationships between policy measures and employment outcomes.

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General equilibrium model

Reference guide on employment impact assessment

Main points

- A general equilibrium model helps account for economy-wide effects of changes brought to an economic system by external factors, such as a policy change.
- A general equilibrium model takes into account the interdependence among all sectors, markets and regions or countries.
- The types of general equilibrium modelling include computable general equilibrium and dynamic stochastic general equilibrium.
- Typically, a general equilibrium model is used before a policy is implemented to simulate the potential economic and social impacts.

Objective of a general equilibrium model[6]

A general equilibrium model, commonly known as GEM, helps to account for economy-wide effects of changes brought to an economic system in its equilibrium state by external factors, such as a policy change. Unlike a partial equilibrium model, which incorporates only a portion of the market, a general equilibrium model factors in the interdependence among all sectors, markets and regions or countries. The analysis of this interdependence allows for the identification of winners and losers resulting from a policy change. Typically, a general equilibrium model is used before a policy is implemented to simulate the potential economic and social impacts. By comparing policy-induced scenarios in which an economic system remains in general equilibrium (that is, the circular flows of income and expenditure are balanced) against a baseline scenario, this method allows for consistent comparative analysis of policy scenarios. This modelling is thus widely used to assess the potential distributional and long-term effects of policy changes, typically trade liberalization, taxation and climate-related policies.

6 Kee Beom Kim authored this section of the reference guide.

▶ How this method works

There are several types of the general equilibrium model, including the computable general equilibrium model, commonly known as CGM, and the dynamic stochastic general equilibrium model, commonly known as DSGM. The most widely used among them are the general equilibrium models, which usually starts by mapping the flow of goods and services and the factors of production and payments in an economy. This mapping is typically in the form of a social accounting matrix.

A computable general equilibrium model is calibrated by fitting the social accounting matrix to a system of parameters and equations that define the behaviour of different economic agents (such as utility maximization of households and profit maximization of firms). If calibrated correctly, the model replicates the base-year equilibrium, which serves as the point of comparison for counterfactual-equilibrium analysis of any hypothetical policy change (Whalley 1985).

A computable general equilibrium model includes static, recursive-dynamic and dynamic versions. In the dynamic model, the way on which economic agents adjust to the shocks over time is taken into account. For example, in relation to the labour market, the supply of labour is usually taken as a given parameter in a static model. But in the dynamic model, assumptions about the labour force participation rate is combined with population forecasts, to determine the supply endogenously (Gibson 2011).

An overlapping generations model is a variant of the computable general equilibrium method, modelling each successive generation. It is used when incorporating the allocation of resources across different generations. The dynamic stochastic general equilibrium model seeks to capture business cycle fluctuations and uncertainty and focuses on shorter-term impacts of policy changes.

▶ Typical questions this method can answer

- **What are the effects of trade liberalization (or alternatively, increased tariffs) on employment and wages?** See, for example, the International Labour Organization and the Asian Development Bank (2015), which used a computable general equilibrium model to assess the potential economic and social impacts from deepening Association of Southeast Asian Nations integration in the form of tariff removals, liberalization of non-tariff barriers and trade facilitation in the form of reduced fixed trade costs.
- **Who gains and who loses from the implementation of climate and energy policies?** See, for example, Chateau, Bibas and Lazi (2018), who used a computable general equilibrium model.
- **What is the effect on employment resulting from reductions in corporate taxes?** See, for example, Alvarez-Martinez et al. (2018), who used a computable general equilibrium model to assess the impact on gross domestic product, tax revenues and employment resulting from reductions in corporate taxes, taking into account cross-country spillovers in corporate taxes and investment.
- **What are the international spillover effects of lower labour costs?** See, for example, Charpe and Kuhn (2015), who used a two-country dynamic stochastic general equilibrium model to assess the international spillovers of a decline in labour costs.
- Although they are not forecasting tools, computable general equilibrium models are also used to inform employment projections. For example, the medium- to long-term employment projections of New Zealand's Ministry of Business, Innovation and Employment used a computable general equilibrium model for their projections (MBIE 2018).

▶ Centrality of employment in the method

Employment is a central variable of interest. But first-generation and standard computable general equilibrium models have often been criticized for not modelling the labour market in detail, including a flexible, market-clearing wage that can lead to full employment. Newer models often incorporate a range of labour market imperfections, including unemployment, various wage-forming mechanisms, dual labour markets and larger numbers of representative households (for a comprehensive review of labour markets in computable general equilibrium models, see Boeters and Savard 2011).

▶ Pros and cons

Pros

- A general equilibrium model has the ability to assess potential economy-wide and second-round effects of policy changes, including their distributional implications and trade-offs between policies.
- Computable general equilibrium models provide a consistent framework, based on microeconomic foundations rather than historical relationship, for quantitative analysis.
- Newer computable general equilibrium models incorporate a range of labour market imperfections, and the minimum data requirements can be limited to one social accounting matrix and one household survey (rather than time series), making these models useful for developing countries.

Cons

- A general equilibrium model may have difficulty to assess what is driving the results (the "black box" argument).
- The choice of the closure rule can impact results.
- There can be uncertainty over parameter values, with some taken from empirical studies of developed countries while similar values for developing countries are often lacking.

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Input–output table and social accounting matrix

Reference guide on employment impact assessment

Main points

- An input–output table and a social accounting matrix help identify changes in production and labour demand.
- An input–output table and a social accounting matrix provide an estimate of the links between different activities and institutional sectors.
- They link how changes in final demand can impact a sector of the economy.
- Despite some potential shortcomings, these are two of the most robust methods for assessing the impact of policies, projects and almost any shock that can be reduced to a monetary impact.

Objective of the input–output table and social accounting matrix approaches[7]

The input–output table, commonly known as an I–O table, and a social accounting matrix, also commonly known as SAM, are the inputs that make it possible to estimate the meso-level employment effects of certain interventions in an economy. Because both approaches contain sector-level information, they are useful for analysing the effects of sector-specific interventions, in particular how such interventions impact other productive sectors and other sectoral institutions in an economy.

The input–output table and social accounting matrix only contain information expressed in monetary values (the transactions between sectors and institutions). Thus, they are typically applied with sectoral-level employment data to estimate the employment effects. They are widely used for different kinds of economic analysis. This reference on these tools does not capture all of their potential applications and instead focuses on their use for analysing employment effects.

7 Gabriel Michelena and Luis Villanueva authored this section of the reference guide.

▶ How these tools work

An input–output table describes the sale and purchase relationships between producers and users of goods and services within an economy and shows the flows of intermediate and final goods and services defined according to industry (sector) outputs for a given year. The social accounting matrix is an extension of the input–output table and adds other flows among different sectoral institutions in the economy, such as transfers from government to households, a broader set of taxes or domestic payments to the rest of the world (monetary flows recorded in the balance of payment).

One of the main advantages of a social accounting matrix is that it combines information from the national accounts, the balance of payments and the input–output table into a single consistent matrix. The main rule is that the sum of the rows and columns must be equal. As for the columns, the total sum at the bottom represents the budget constraint faced by each sector (the total expenditure by sector). In this sense, there are no black holes – every purchase necessarily implies a sale, and every payment necessarily implies an income. As Godley and Lavoie (2007) explained: "Everything must come from somewhere and go somewhere."

One of the main applications of the input–output table and social accounting matrix is to construct an economic model that permits the construction of counterfactual scenarios and to link variations in final demand to potential changes in output and employment. This basic model uses the so-called Leontief inverse matrix as its cornerstone. The Leontief inverse, also known as the "total requirements matrix", represents the direct and indirect requirements of output in each sector that is produced to satisfy a unit of final demand.

The sum of all the elements of a single column in this matrix yields what are called "output multipliers". The output multiplier represents the total amount of output (in some currency unit) required to satisfy a unit of final demand (in the same currency unit). In an extreme case, where an activity requires no intermediate inputs, the multiplier is equal to 1. In a more general case, the initial effect of an exogenous demand shock in a given sector is amplified by the internal flows of all activities that are directly and indirectly affected, and the multiplier will be greater than 1.

The multipliers derived from the total requirements matrix can be used to assess the direct and indirect employment effects, using the corresponding labour coefficients (labour divided by the initial level of output). A typical example could be an initial increase in final demand of construction services due to augmented government spending on infrastructure. The increased output of the construction sector would in turn also trigger a raise in the demand from other sectors that supply inputs to it, such as cement, gravel and steel. Using the output multipliers and the labour coefficients, these output effects are then used to estimate the corresponding impacts on employment demand.

The Leontief inverse is known as a type I matrix and does not include any feedback effect from the final demand. Therefore, the multipliers obtained from this matrix are also regarded as type I multipliers. However, this matrix can be extended to endogenize any component of the final demand. The four possibilities available are: consumption, investment, government spending or exports.

In the so-called Leontief type II matrix, household consumption is usually removed from the final demand vector and is endogenized by incorporating it into the original transaction matrix to capture the induced impact of a change in final demand. In economic terms, the induced effect captures the fall or increase in output resulting from the new expenditure that households will make, based on the change in their income. Thus, with an increase in final demand, sectoral output will grow, and the wage bill will increase due to the rise in employment, which in turn will have a positive impact on consumption.

Additional considerations

Depending on the availability of data, the levels of sectoral aggregation represented in an input–output table and social accounting matrix may vary. Both approaches are considered a "meso" model, wherein the economy is analysed at a sectoral level of aggregation, such as crops, food, textiles, construction or financial services. On the other hand, "macro" models, such as the dynamic stochastic general equilibrium (DSGE) or the stock-flow consistent model (SFC), usually work with large economic aggregates, such as consumption, investment or GDP, while "micro" models are based on the level of individual units, using highly disaggregated micro data (Bourguignon and Spadaro 2005). If sectors are highly aggregated in the input–output table or social accounting matrix and a specific subsector is of special interest, a sub-activity account can be created, and the effects of this new sector can be estimated. This exercise typically requires using external information from other data sources in addition to the input–output table and social accounting matrix, such as a household or employment survey, to identify the employment or the output in those more disaggregated activities.

It is important to be aware of assumptions inherent in the use of input–output tables and social accounting matrices when conducting this multiplier analysis, such as:

- There are no material, labour, foreign exchange or capital supply constraints in the economy. If demand increases, the economy will match this shock with a greater supply of goods and services. This also means that there is no change in prices as a result of the adjustment in demanded quantities.
- Technology in the production process is assumed constant, and the ratio of different inputs to output (usually known as "technical coefficients") remains the same, even if output or demand is shifting in the simulations.

The extent to which these assumptions hold in the context being analysed should be determined. The input–output table and social accounting matrix approach is usually recommended when carrying out only short-run analyses, in which the economic structure and technology remain unaltered. Fixed-labour coefficients may lead to overestimation of employment outcomes due to the high value of labour coefficients in the agriculture sector (especially in developing countries). The assumptions behind the input–output table and social accounting matrix multiplier analysis also constrain the causal structure of the model and do not allow for the investigation of more complex causal relations, such as capacity changes on the supply side and non-linear effects.

▶ Typical questions this method can answer

- What are the employment effects of investments, most prominently in infrastructure, by governments or donors?

  See, for example: Luis Villanueva and Keren Neza, Employment Impact Assessment of the Malawi M1 Road Rehabilitation Project, ILO, 2023.

- What are the employment effects of increasing exports of products from certain sectors?

  See, for example: ILO, Trade, Investment and Employment in the Southern Mediterranean Countries, 2022.

- How the employment impacts of increased outputs differ according to the various types of effects.?

  See, for example: ILO, A Brief Review of Input–Output Approaches on employment impact assessment, 2024.

- Which sectors have the highest job-generation potential?

  See, for example: Pamphile Sossa and Mbock Pierre Olivier, Employment impact assessment of the 11th European Development Fund to the National Participatory Development Programme in Cameroon, ILO, 2024.

Centrality of employment in the method

The information on employment in the input–output table is usually limited to the share of wages in the production process. The social accounting matrix also contains information on the distribution of wage payments to households. To estimate effects on employment, it is necessary to convert payments and outputs, which are in monetary values, into employment (hours worked, number of workers or jobs). This is typically done in one of two ways:

- Additional output is converted into additional employment using employment-output ratios (total employment in sector X and total output of sector X).
- Factor payments to labour (total wages) are divided by the average wages in the sector to obtain employment in the sector.

Neither an input–output table nor a social accounting matrix contains an explicit modelling of the labour market. Yet, the lack of supply constraints can be interpreted as a labour market in which demand is a function of the level of output and the supply curve is horizontal. It is thus assumed that the labour demand is always able to meet the level of unemployed workers in the market who are ready to work.

Pros and cons

Pros

- The input–output table and social accounting matrix are relatively simple to use.
- They are available for most countries and thus widely applicable. And they are the basis for more advanced modelling methods, such as computable general equilibrium models.
- The underlying assumptions are easy to understand.
- When the necessary information is available, both methods are useful for analysing other non-employment-related effects (such as CO_2 emissions and water requirements) and to understand the main interactions in an economic structure.

Cons

- The analysis and the results provided are usually static.
- Because the construction of an input–output table and social accounting matrix is resource-intensive work, some of the data sets are usually outdated, especially in developing countries.
- Due to some of the assumptions implicit in the methodology, in particular that there are no supply constraints and that there are fixed technical coefficients (technological relationships), the results tend to overestimate the employment effects.
- In some cases, the sectors included in the matrix can be highly aggregated, with a wide variety of economic sub-activities aggregated in a single sector, which could distort results if the specific intervention analysed is very specific and delimited.
- The methodology provides little insight into the quality of jobs created; typically, it is assumed that jobs created will be, on average, similar to the employment structure in a sector as a whole, so firms will not change the composition.
- The Leontief model is dual, so the price system is assumed to be independent of the quantity model. This feature means that price and demand shocks cannot be analysed simultaneously.

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Structural model for sustainable development

Reference guide on employment impact assessment

Main points

- The structuralist models, such as the structural model for sustainable development, are simulation analysis tools based on a sector-disaggregated, economy-wide model.
- The method captures the structure of the economy, its institutions, social actors and their economic relationships.
- The method uses a social accounting matrix and other socio-economic data and can be adjusted to different social accounting matrix structures.
- The specific structural model for sustainable development provides richness, flexibility and adaptability, but it requires a computing platform and knowledge of the underlying model.
- The model inherently focuses on employment, income distribution and redistribution and can provide insights on employment quality.
- The model strikes a balance between the need for rich description of the economy and the need to provide clear insights on the results.

Objective of the structural model for sustainable development[8]

The structural model for sustainable development, also known as SMSD, is quantitative simulation tool for assessing the sector-disaggregated, economy-wide impacts of interventions, such as policies or specific investments. The model captures the impact of policies on macroeconomic dimensions while providing sectoral details. These include effects on different products, activities, labour types (age group, gender, relative wages or earnings, employment and formality status, skills and occupations), households (by income levels and rural versus urban) and other institutions. The data can be captured at different levels of aggregation, depending on the nature of the intervention and of the analysis.

8 Massimiliano La Marca and Xiao Jiang authored this section of the reference guide.

The tool is based on a "structuralist simulation model". The structuralist method focuses on the structure of the economy, the role of its institutions, the socio-economic actors and their economic relationships. It allows assessing how policy interventions propagate throughout an entire economy, sectors and groups of population and their interactions with the existing and persistent institutional and economic structures as a system of interlinked socio-economic groups. Assessing such a range of outcomes contributes to better evaluating interventions and helps to inform and strengthen the consistency of policies and interventions aimed at achieving simultaneously a variety of developmental objectives.

The method and, in particular, the structural model for sustainable development, can be used for different types of economic analysis and their social and environmental extensions. This reference guide does not capture all of the potential applications of the model but instead focuses on the use of the tool for the assessment of employment impacts.

▶ How the tool works

Like the input–output multiplier analysis, the structural model includes interindustry linkages and, like the social accounting matrix multiplier analysis, it extends them to production, income, distribution and consumption linkages. It additionally includes price and quantity interaction, which can take into account more complex distributional behaviours and effects in production, consumption and investment. Contrary to standard computational general equilibrium models, the price mechanism and the assumption of competitive markets and full (or fixed) employment of resources do not dominate the economic adjustment mechanism nor the results of the analysis.

The structuralist approach emphasizes the relationship between the structure of the underlying economy and the impacts of policies on social and economic outcomes.[9] The structural model highlights how policy interventions are affected by existing and persistent institutional and economic structures, such as a given production capacity and fixed production structures in the short run, non-price-taking behaviour in the product and factor markets and persistent patterns of expenditure and savings based on income groups. The changes in the capacity constraints, often induced by the policy or investment to be assessed, generate rich medium- to long-term effects across an economy on production, prices, employment and incomes, which thus provide some characteristic analytical advantages to this methodology over computational general equilibrium models or input–output and social accounting matrix multiplier analysis.

Developed by the International Labour Organization, the tool is characterized by a high degree of flexibility because it can include various country- and area-specific modules. For instance, the core model can be extended by adding differentiated labour market and formal versus informal economy behaviour. It can include macro and micro simulation interactions to carry out more detailed and targeted analyses looking at social as well as environmental sustainability. The tool has been applied to a variety of countries and projects.

Because the structural model is a quantitative assessment tool, data availability is key. The essential underlying database is a country-level social accounting matrix.[10] Such a method records the transactions among the main actors in an economy over a period of time. Transactions are represented in a set of interlinked accounts that record incoming and outgoing resources of such actors. Hence, a social accounting matrix links together the macro statistics of national accounts with the micro statistics of the

9 For the origin of this approach and modeling framework see Taylor 1983 and 2016.

10 See ILO 2017 for the social accounting matrix structure and analysis and relationship to the input–output analysis.

labour market, household income and consumption and other social statistics (European Commission 2003). Ideally, each country-level assessment will initially require careful work on data collection from various sources, on processing and on compilation of the social accounting matrix. Some support from the national statistics office is typically required. Fortunately, thanks to the flexibility of the framework, the structural model can also adapt to existing social accounting matrices, given its analytical power will be limited by the structure and details of the existing social accounting matrix.

The model will need clear, well-specified and quantifiable policies and interventions to be translated into model inputs and scenarios to simulate. Hence, the analysts using the structural model must work closely with relevant stakeholders to identify the interventions to be assessed and to translate them into model inputs. The results of the model also require careful interpretation due to the data quality, model assumptions and the selected adjustment mechanisms. These latter two activities will imply sharing all information on the model methodology as well as undertaking some knowledge transfer.

The structural model relies on some assumptions:

- The social and economic structure, with its institutional behaviours and market structure, is a persistent feature of an economy and reproduces itself in absence of major policy or structural changes. Such structure is largely represented by a country-level social accounting matrix. This implies that some production and consumption coefficients are fixed in the short term because investments or policy interventions are needed for production and consumption structures to change.

- Production sectors can be grouped in different categories because they are characterized by different adjustment mechanisms. For example, agriculture, energy and other utility sectors can be capacity-constrained due to lack fertile land or large capital investments. Their domestic supply has an upper bound in the short term, and demand can generate price adjustment and/or import effects that affect the entire economy.

- Investment operations can build up capacity in these sectors to induce “capacity effects”, which ultimately push an economy to a higher growth path that generates long-term employment impacts.

- Many sectors that are not constrained by capacity can adjust to (limited) increase demand. The assumption is that firms in these sectors have a certain excess capacity due to the existence of a large pool of unemployed and underemployed labour and/or capital. In such a situation, firms can set their supply price mark-up at more than their cost of production.

- While some sectors are more likely to operate under full capacity than others, the structural model allows for an endogenous determination of the state of the sector. The relationship between demand and capacity has an important role in the sectoral and economy-wide outcomes. This underlines the role of investment and demand policies.

- In sectors where capacity constraints are not binding and mark-up pricing dominates, production costs are directly reflected onto product prices. Because labour and intermediate consumption of products are part of the cost of production and coefficients are fixed in the short term, any change in wages or in the price of intermediate consumption products affects the supply price. Thus, price increases propagate across the economy via cost-push.

- For the baseline model, labour supply is assumed to be unlimited at an exogenously determined nominal wage rate. Further assuming fixed productivity, employment and income are generated by additional levels of domestic industrial activity, which depends on the demand for domestic products and the product mix of each industry. Demand in this model depends on prices and the distribution of income. Tax rates are assumed to be fixed, so government budget balance adjusts in response to an intervention. The current account adjusts freely, assuming a fixed nominal exchange rate, whereas the real exchange rate still adjusts with price. In response to an intervention, the saving–investment balance is restored, with incomes adjusting to generate the right amount of savings to meet investment.

Some of these assumptions are more likely to characterize developing economies, which makes the structural model a fit for development and sustainability analysis. Thanks to the flexibility, some of the baseline assumptions and adjustment mechanisms can be altered to fit the purpose of the analysis. For example, the assumption of fixed labour productivity can be relaxed to take into account productivity growth, and more complex labour market adjustment mechanisms can be implemented with the given data and knowledge.

▶ Typical questions this tool can answer

- What are the social and economic impacts of an investment operation in a specific sector or a group of sectors?[11] What kind (by gender, age group, formal or informal status) of employment is the investment (or a set of investments) generating?
- What is the impact on household income levels and inequality?[12]
- What are social and economic impacts of social protection policies? This can cover expansion of social transfers, extension of coverage under a contributory and non-contributory scheme, formalization of informal sectors and implementation of universal basic income.
- What are alternative financing mechanisms and their implications for the macroeconomy?
- How do employment effects of increasing outputs differ between sectors?
- What are the complementary policies needed to make transitions truly just? For instance, what are the impacts of specific measures facilitating energy transition on labour and incomes, and what is the nature and size of the complementary industrial, employment and social protection policies that will enable the transitions and offset the potential negative socio-economic impacts?

The policy questions that can be addressed will depend on data availability and country-specific circumstances.

▶ Centrality of employment in the tool

Employment is central to the underlying methodology used by the structural model for sustainable development. A social accounting matrix is constructed with a full set of accounts that trace income generation, distribution and redistribution, as well as a detailed employment satellite account. When the social accounting matrix is compiled following structural model guidance, the compensation to employees block should contain detailed information on factor-income flow from activities to employment by occupations and formality status. The factor incomes are distributed to various institutions and households by income quintiles or deciles as well as by rural and urban. In this case, the structure of the labour market is embedded in the model from the beginning.

11 Examples of impacts include but are not limited to: indicators for real growth, inflation, employment, government budget, income distribution, poverty and income equality.

12 An example of infrastructure investment impact analysis can be found in Jiang and La Marca 2023.

Detailed employment satellite accounts are often carefully constructed based on data from labour force surveys, which often include information on employment by gender, age and other characteristics. Hence, the impact assessment can tap into the multifaceted qualitative dimensions employment.

The adjustment mechanisms behind the labour market in the baseline model are intentionally simplified in the base version of the model, thus leaving room for more complex, comprehensive and economy-specific mechanisms when needed.

▶ Pros and cons

Pros

- The structural model for sustainable development strikes the balance between the complexity needed to capture real economy phenomena and the parsimony needed for its applications and the interpretation of results.
- Because it is a full macroeconomic simulation model, with extensive price and quantity interactions, it is potentially more realistic and able to answer more complex questions, when compared to the input–output and social accounting matrix multiplier analyses.
- It is flexible enough to adapt to any social accounting matrix, including publicly available ones, and to address a range of policy and development questions.
- It is a tool that puts employment and job quality, social and environmental effects and policy consistency on centre stage.
- It allows the construction of user-friendly interfaces that suit the capacity of the end-users.

Cons

- The application and comprehensiveness of the structural model and the scope of analysis depend on the quality of the underlying data. Thus, data processing and production can be a demanding task for more complex analyses.
- As with other macro and sector-disaggregated models, technological change is difficult to estimate and to incorporate without relying on strong assumptions. There can be uncertainty over some estimated parameters, and sensitivity analysis may be needed.
- The sectors included in social accounting matrices can be highly aggregated, with a variety of economic sub-activities aggregated in a single sector. This can bias the results if the analysis is on specific sectoral interventions and no further disaggregation is possible.
- Because it is a quantitative simulation tool, it requires interventions (a policy or investment) to be fully quantified and translated into an analytical scenario that can be included in the model.
- The analytical structure of the model shares some elements with the input–output and social accounting matrix analyses as well as with the computable general equilibrium models. Yet, the full model and its behaviour depart from those standard approaches, and the interpretation of its results requires some model-specific knowledge.

- The model requires a computing platform, and it is currently available in Wolfram Mathematica language. Computing efficiency and accuracy will depend on the programming and software used.

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Growth decomposition

Reference guide on employment impact assessment

Main points

- Growth accounting decomposition leads to deeper understanding of the structural dynamics of economic development.
- Growth accounting decomposition helps determine sectoral drivers of productivity growth.
- Growth accounting decomposition can address structural dynamics of gender differences in employment growth.
- Growth accounting decomposition helps determine the relative importance of within-sector versus cross-sector (reallocation) effects on productivity growth.

Objective of growth decomposition[13]

Recent economic policy discussions have been marked by a renewed interest in structural transformation. One reason for this is concern over the potentially negative implications of deindustrialization in developed and developing countries and, in particular, "premature" deindustrialization among the latter. Another is the debate on whether advanced services relying on information and communication technology can function as a leading sector in economic development, characterized by static and dynamic economies of scale and positive spillover effects. The debate largely revolves around whether and how advanced services can substitute for or, rather, complement manufacturing – the archetypal leading sector.

A thick strand of the literature on structural transformation focuses on the reallocation of labour from lower- to higher-productivity sectors and the accompanying changes in productivity within sectors. These are referred to as, respectively, employment reallocation and within-sector effects on aggregate productivity growth.

Several empirical studies have used growth accounting decomposition to evaluate the relative importance of employment reallocation and within-sector effects as well as the sectoral drivers of

13 David Kucera authored this section of the reference guide.

aggregate productivity growth, particularly labour productivity growth (see Kucera and Jiang 2018; Dasgupta, Kim and Caro 2017; Kucera and Roncolato 2016; Roncolato and Kucera 2014; Ocampo, Rada and Taylor 2009; Timmer and de Vries 2009; Pieper 2000). As Ocampo, Rada and Taylor observed (2009: 42), labour productivity growth has a central role in economic development: "Historically, labor productivity increases have been the major contributing factor to growth in real GDP per capita."

From the employment perspective, structural transformation is thus typically considered to have profound implications through compositional shifts towards leading sectors, as well as corresponding changes in per capita incomes and aggregate demand via the Kaldor–Verdoorn growth dynamics (Kaldor 1967; Verdoorn 1949).

▶ How this method works

There are different accounting decomposition methods, which have different pros and cons, depending on the measure of change being decomposed. The method presented here decomposes labour productivity growth. Other methods decompose differences in productivity levels (for example, what McMillan and Rodrik 2011 used).

▶ Typical questions this method can answer

Growth accounting decomposition methods can reveal patterns that are not otherwise readily apparent and can therefore deepen the understanding of processes of structural transformation. Kaldor (1968) hypothesized that employment reallocation effects are generally less important than the within-sector effects induced by employment reallocation, for which growth accounting decomposition methods provide a test. These methods can also usefully inform policy, insofar as they facilitate learning from the examples of more and less successful countries in terms of economic development. For example, the view that differences in employment reallocation distinguish rapidly growing economies has led some analysis to search for barriers to labour mobility across sectors (McMillan and Rodrik 2011).

Growth accounting decomposition methods can usefully inform sectoral policies by clarifying whether and how sectoral drivers within more and less successful countries differ from each other. More specifically, the methods can answer whether aggregate labour productivity growth in more-rapidly growing economies tends to be driven by a particular set of sectors and, if so, which ones? These methods have also been used to study changes in female shares of employment, in particular to identify the sectoral drivers of these changes and the relative importance of employment reallocation and within-sector effects, so as to better understand gender aspects of structural transformation (Esquivel forthcoming; Tejani and Kucera 2021). Some of these studies focus on better understanding the widely observed pattern of rising female shares of manufacturing employment in the early stages of export-oriented industrialization, often followed by falling shares (Kucera and Tejani 2014).

▶ Centrality of employment in this method

Pieper (2000) and Ocampo, Rada and Taylor (2009) provided useful decompositions of labour productivity growth. Aggregate labour productivity is defined as total value added over total employment, or , and sectoral labour productivity is correspondingly defined as . Aggregate labour productivity can be expressed as:

$q = X/L = \text{å}x_i/\text{å}l_i$ [Eq. 1]

Taking first-order differences with respect to time , labour productivity growth can be expressed as:

$\text{x} = \text{å}[\theta_{i0}\,(g_i\text{-}n_i) + (\theta_{i0}\text{-}(q_1/q_0)l_{i0})n_i]$ [Eq. 2]

where:

$\text{x} = (q_1\text{-}q_0)/q_0$

$n_i = (l_{i1}\text{-}l_{i0})/l_{i0}$

$g_i = (x_{i1}\text{-}x_{i0})/x_{i0}$

$q_{i0} = x_{i0}/X_0$

$l_{i0} = l_{i0}/L_{i0}$

The within-sector effect on labour productivity growth is represented by the left-hand bracketed term in equation [2], that is:

$\text{x}_w = \text{å}[\theta_{i0}\,(g_i\text{-}n_i)]$ [Eq. 3]

In other words, the within-sector effect is the difference between sectoral value-added growth and employment growth weighted by the output share of the sector. Positive within-sector effects result when sectoral value-added grows faster than sectoral employment.

The interaction term is represented by , or the result of first-order differencing in discrete time steps, while the interaction effect is the difference between aggregate labour productivity growth and the sum of within-sector and reallocation effects. Leaving the interaction term aside, the employment reallocation effect on labour productivity growth is represented by the right-hand bracketed term in equation [2], that is:

$\text{x}_r = \text{å}[(\theta_{i0}\text{-l}_{i0})n_i]$ [Eq. 4]

The reallocation effect is the difference between sectoral output and employment shares multiplied by sectoral employment growth. Positive reallocation effects result when sectoral employment grows in sectors for which the difference between sectoral output and employment shares is positive, that is in sectors with above-average labour productivity.

Leaving aside the interaction effect, each sector's contribution to aggregate labour productivity growth is its labour productivity growth weighted by its relative output – the within-sector effect – plus its employment growth weighted by its relative labour productivity – the reallocation effect. Relatively larger sectors will tend to contribute more to aggregate labour productivity growth through the within-sector effect. The larger the difference in labour productivity among sectors, the larger the potential increases in aggregate labour productivity through reallocation effects, provided that employment shifts from less- to more-productive sectors.

▶ Pros and cons

Pros

- Economic development and structural transformation is generally thought of in terms of growth and not differences in levels. This method helps to understand what drives growth.
- Unlike most employment impact assessments, results from growth accounting decomposition are not estimates but rather tautologies.

Cons

- The limitation of looking at differences in productivity levels is that high-income countries with high productivity levels have greater differences in productivity levels for a given rate of productivity growth. Thus, results from the decomposition of differences in levels of productivity growth do not lend themselves as readily to comparison among countries and can create the false impression that higher-income countries are undergoing more rapid structural transformation than lower-income countries.

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Local multiplier

Reference guide on employment impact assessment

Main points

- The local multiplier method is well suited for meso-level (regional, subregional, city) impact assessments. By using local data rather than national (aggregate) data, they capture local effects of investments, loans, infrastructure projects or regional policies.
- The local multiplier approach helps identify industrial sectors with high employment potential, especially related to local-level interventions.
- Advantages of the local multiplier method is its focus on employment and that it can account for productivity growth, providing an accurate estimation of employment effects.
- Local multipliers can be decomposed by gender, age group, skill level and employment status, thus providing a detailed understanding of the employment impact. They are dynamic over time, relying on at least two consecutive population censuses.

Objective of the local multiplier[14]

A local multiplier is an empirical methodology that aims at producing sectoral employment multipliers (Moretti 2010). The local multiplier answers the following question: How many jobs are created in the (local) economy when one job is created in sector X in the same municipality?

This approach is useful for policymakers who want to determine the industrial sectors with high employment potential, which is often the case when designing national employment policies. This approach also informs policymakers on the specialization and dynamics of local labour markets, measured at the level of municipality, urban agglomeration or county. It is an important method to track structural transformation at the subregional level. Additionally, local multipliers can be used to measure the indirect and induced employment effects associated with development programmes or financial investment from development institutions.

14 Matthieu Charpe authored this section of the reference guide.

▶ How this method works

The local multiplier effect refers to the overall impact on a local economy resulting from an increase in local labour demand within the tradable sector. The starting point is a competitive economy producing both tradable and non-tradable goods. Tradable goods are consumed nationally and can be produced in different locations if cost inflation becomes too high. Producers of tradable goods are price takers while non-tradable goods are produced and consumed locally, leading to local pricing.

The shock considered is a permanent increase in local labour demand within a subset of the tradable sector. This increase could be due to local authorities attracting new firms (Adams 2016) or an increase in the productivity of existing jobs. Applications have also considered the impact of an increase in public employment (Lee, Ko and Kim 2024; Becker, Heblich and Sturm 2021; Jafre-Monseny, Silva and Vazquez-Grenno 2020; Auricchio et al. 2019; Faggio and Overman 2014), high-tech firms (Goos, Koninas and Vandewever 2018) or natural resource boom (Munasib and Rickman 2015). The methodology aims to measure the impact of this shock on the local economy, including both the non-tradable sector and the rest of the tradable sector.

This approach considers the following transmission channels on the non-tradable sector:

- **Consumption of locally produced goods:** Employment in the non-tradable sector will increase as new workers in the tradable sector spend some of their earnings on locally produced goods and services. The magnitude of this impact is influenced by the demand for non-tradable goods and the skill levels and wages associated with the new tradable jobs.
- **Technology and labour intensity in the non-tradable sector**: The number of jobs created in the non-tradable sector reflects labour productivity (and its change over time) as well as the level of wages.
- **Local supply chain**: The characteristics of the local supply chain determine how much local non-tradable goods are consumed due to the increase in tradable employment.
- **Price and wage inflation**: Negative effects centre around price and wage inflation, which are (partly) determined by the elasticity of labour supply and labour mobility.

The impact on other tradable sectors is driven by two opposite effects: (i) the crowding out of other tradable firms due to wage inflation; and (ii) the agglomeration effects that capture the benefits to firms when located near one another.

Empirically, the local multiplier is an easy-to-use and intuitive approach that makes use of the geographic, employment and industry information contained in the population and housing census.

The starting point of this approach is to construct a cross-section or panel database consisting of sectoral employment measured at the local level over time. The cross-section dimension is given by the spatial information from measures at the level of municipality, urban agglomeration or county contained in the census data. Employment is measured at the level of industry, differentiating the manufacturing, services and agriculture sectors.

The impact of the creation of one job in location m and subsector j on another subsector k is estimated via a reduced form equation. The size of the effect reflects the different transmission channels noted previously, such as the demand for local goods, labour intensity and local supply chains, as well as wage and price inflation.

The following table summarizes the local multiplier of tradable jobs on non-tradable jobs, as estimated in the literature for a selection of high-income and low-income countries. The table also highlights the impact of creating non-tradable jobs on tradable jobs. Local multipliers usually deliver an intuitive hierarchy of sectors, with the highest employment potential in manufacturing, followed by services and agriculture. In a high-income country, the creation of one manufacturing job is usually associated

with one job created in services in the United States and 0.5 jobs in Sweden. In sub-Saharan Africa, the multiplier is up to five times larger than in the United States. The main reason is related to the wage differentials between manufacturing and services jobs. An additional result is that the size of the multiplier increases with skills as well as with the degree of formality in the employment relationship (not shown here). Contrastingly, non-tradable job creation has little impact on tradable jobs.

▶ Selection of local multipliers between tradable and non-tradable jobs

Country	Author	Tradable on non-tradable	Non-tradable on tradable
United States	Moretti (2010)	1.59***	
	Dijk (2017)	1.02***	
Sweden	Moretti and Thulin (2013)	0.48*	
Benin	Charpe (2022)	1.34***	0.05*
Ivory Coast		3.01***	0.18***
Ghana		1.61***	0.26***
Malawi		1.78***	0.18***
Mali		1.06***	0.11***
Mozambique		2.27***	0.18***
Rwanda		4.62***	0.12***
South Africa		0.53**	0.19**
Tanzania		3.75***	0.08***
Zambia		3.68***	0.14***

Note: Examples of local multipliers between tradable and non-tradable jobs as estimated in the literature. *= statistically significant at 10%; **= statistically significant at 5%; ***= statistically significant at 1%.

Source: Charpe 2022; Osman and Kemeny 2022; Osei and Sengupta 2019; van Dijk 2017; Moretti and Thulin 2013; Moretti 2010.

▶ Typical questions this method can answer

- What are the industrial sectors with the most promising employment potential?
- What is the sectoral employment distribution at the local level, such as at the municipality level? And how has it evolved over time?
- How many jobs are created in the (local) economy when one job is created in sector X in the same municipality?
- How does the quality of jobs that are created impact the local labour market?

▶ Centrality of employment in the method

- One of the main advantages of the local multiplier method is the centrality of the employment dimension. This allows the local multiplier to account for productivity growth, thus leading to accurate estimations of the employment effects (Bartik and Sotherland 2019).
- Local multipliers are dynamic over time because they rely on at least two consecutive population censuses, factoring in the changing sectoral patterns often at work in low-income countries.
- The wealth of data given by the number of administrative entities considered allows the possibility to estimate the multiplier and to produce confidence intervals.
- Local multipliers can easily be decomposed by gender, age group, skill level and employment status (formal and informal). The employment module contained in the population census informs not only on employment but related dimensions, such as industries or employment status. The broader information of the census data on age, gender and educational attainment also opens up the possibility to measure sectoral multipliers for all or a combination of these subgroups.
- The local multiplier approach is well suited for the meso level: regional, subregional and city-level impact evaluation. This approach is meant to capture local effects, for instance, the impact of a local investment, a new loan to a firm, an infrastructure project or even regional policies. The use of local data improves precision with respect to the use of national data.

▶ Pros and cons

Pros

- The local multiplier method is well suited to the context of low- and middle-income countries, which are characterized by a scarcity of data, especially when it comes to national accounts. (The census offers the primary data collected in low-income countries.)
- The large number of data points and the time dynamic behind the estimation are useful.
- The method provides a picture of the local employment dynamic and specialization.
- There are no preconceived assumptions on the sign and the size of the multipliers, which can be positive, zero or negative.
- It relies on an intuitive hierarchy of sectors: manufacturing, then services and then agriculture.
- It offers local analysis that fits well with the type of intervention financed by United Nations agencies or international finance institutions.

Cons

- There is a challenge associated with the instrumental variable in low-income countries with homogenous industrial specialization.
- The importance of internal migration in low-income countries can bias the effects upwards.
- There is limited sectoral disaggregation, given the geographic disaggregation.

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Sector identification

Reference guide on employment impact assessment

Main points

- This approach, which relies on any of three methods, helps countries identify promising sectors for export growth and structural transformation.
- The approach uses readily-available, detailed export data that can inform a country's process of "self-discovery" by suggesting possible paths for economic development that may not be otherwise apparent.
- By identifying promising sectors, the approach can provide targets for trade, industrial and structural transformation policies.

Objective of sector identification[15]

The objective of sector identification is to provide concrete targets at the sector and product levels to guide trade, industrial and structural transformation policies. By relying on data-driven analysis, these methods can provide transparent criteria to inform the discussion and design of these policies.

Three prominent methods can be used to identify potentially promising products and corresponding sectors for export expansion and structural transformation for any given country: (i) the growth identification and facilitation framework (Lin 2012); (ii) the economic complexity and product space method (Hausmann et al. (2014; Hidalgo and Hausmann 2009; Hausmann and Klinger 2006) and (iii) the International Trade Centre's export potential and product diversification indicators (Decreux and Spies 2016).

The three methods have comparative strengths, albeit to varying degrees.

15 David Kucera authored this section of the reference guide.

▶ How the methods work

The growth identification and facilitation framework is a method to determine potentially promising sectors by purportedly following comparative advantage based on a country's factor endowments. The first step is to identify countries that either have per capita incomes two to four times as high as the country of interest or that had a similar per capita income as the country of interest 20 years ago and that also had average annual gross domestic product growth rates of greater than 6 per cent over those 20 years. These measures provide aspirational benchmarks for the country of interest. The next step is to analyse the benchmark countries' export data at the four-digit Standard International Trade Classification (SITC) level, but asymmetrically, depending on whether the country of interest and the benchmark countries are classified as resource-poor or resource-rich. In other words, look at which sectors have moved into or fallen out of the top-ten ranking of a benchmark country's export shares over 20 years, which can suggest export market opportunities for the country of interest. This list of sectors is reduced by screening the country of interest's specific characteristics but also augmented based on the sectors in that country for which an index of so-called "revealed comparative advantage" has increased in recent years.

The economic complexity and product space method identifies potentially promising sectors largely based on multidimensional analyses of export data at the four-digit SITC level, encompassing roughly 1,000 products. Hausmann et al. (2014), for example, explicitly contrasted their approach with comparative advantage based on factor endowments. The notion of economic complexity was based on the inverse relationship across countries between the diversity and ubiquity of their exports, in which diversity and ubiquity were described as "indirect measures of the capabilities of each country" (Hausmann et al. 2014: 10).

Economic complexity is complemented by the notion of "product space", determined by a mapping of products in which their "proximity" is meant to indicate the extent of similarity in the knowledge required to produce them. As with economic complexity, the extent of similarity in productive knowledge is measured indirectly, based on the probability of any two products being co-exported by countries. Complementing the measures of economic complexity and distance in product space is a measure reflecting the extent of connectedness in product space. For a country, the opportunities for diversification into new products are greater the closer that country's current products are to prospective products in the product space.

From the notion of connectedness, measures of opportunity gain are derived, which for any given country reflects the potential benefit of diversifying into each product that it does not yet successfully export. Hausmann et al. (2014) applied the measures of complexity, distance and opportunity gain to identify potentially promising sectors to promote diversification and economic growth for a country while bringing in product demand, as indicated by the size of global export markets for these products.

The International Trade Centre developed two indicators to determine potentially promising sectors: the export potential indicator and the product diversification indicator. Both are based on export data at the six-digit Harmonized System classification level and encompassing roughly 4,000 products. For any given country and product, the export potential indicator is constructed as the multiplicative product of three components: supply, demand and ease of trade. The central variable for supply is projected market share, measured as the ratio of a country's exports of a given product relative to world exports of that product, adjusted by the projected gross domestic product growth of the exporting country. The central variable for demand is projected imports of a given product into the importing country. Ease of trade is a ratio of actual exports from the exporting country to the importing country relative to hypothetical exports, if the exporting country had the same market share in the importing country as in the world market.

In contrast with the export potential indicator, the product diversification indicator is intended to help countries identify export opportunities in new rather than established sectors. Like the export potential indicator, the product diversification indicator is constructed as the multiplicative product of supply,

demand and ease of trade. Although the demand and ease of trade components are identical in the export potential indicator and the product diversification indicator, supply in the product diversification indicator is based on the economic complexity and product space method described here.

▶ Typical questions these methods can answer

All three methods are designed to inform targeted industrial policies and sectoral strategies. The growth identification and facilitation framework, the export potential indicator and the product diversification indicator methods are meant to determine sectors with export potential for a country. Although the economic complexity and product space method is based on export data, Hausmann et al. (2014) argued, with evidence, that the method is not about export-oriented growth per se but, rather, which sectors can contribute most to economic growth more generally.

▶ Centrality of employment in the methods

The difference between the export potential indicator and actual exports for any given sector in a country of interest provides measures of "untapped" export potential. The export potential indicator and the untapped export potential are the only measures expressed in value terms. Because untapped export potential is expressed in value terms, it can be used to estimate the number of jobs that would be created if a country were to reach its export potential, which the International Trade Centre did in a study by applying fixed employment multiplier analysis to a social accounting matrix for Jordan (ITC 2018). The concern about the potential trade-off between more and better jobs was motivation for Hausmann et al. (2014) to construct separate sector rankings for what they referred to as "parsimonious transformation" versus "strategic bets", based on giving different weights to the measures of complexity, distance and opportunity gain. For developing countries more reliant on agriculture, the parsimonious transformation ranking generally gives greater emphasis to labour-intensive sectors that can more readily provide jobs for low-skilled rural workers.

▶ Pros and cons

Pros

- The economic complexity and product space and the International Trade Centre methods make innovative use of readily available, detailed trade data to provide new insights beyond the realm of trade and into economic development and structural transformation more broadly.
- The results of the economic complexity and product space and the International Trade Centre methods vary by gradations and thus can generate a long list of ranked sectors, which is useful in providing policymakers with a range of prioritized options.

- The results of the International Trade Centre method can be easily translated into employment impacts.

Cons

- All three methods are primarily based on trade data. This means that non-tradable sectors are excluded by construction. Given current data limitations on tradable services in the data sources used by these methods, these sectors are also excluded.
- The measures in the economic complexity and product space method, including the product diversification indicator, are meant to reflect underlying capabilities and productive knowledge – yet, these are only addressed indirectly and abstractly.
- The successful exporting of products depends not just on productive knowledge but on marketing knowledge and production and other costs, including transport costs.
- The growth identification and facilitation framework method is based on thresholds and dichotomies, for which a product either makes the list of potentially promising sectors or does not.

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Skills anticipation

Reference guide on employment impact assessment

Main points

- Employment impact assessments that traditionally focused on the number of employed persons can enhance their scope by incorporating skills requirements based on occupational data. Determining workforce needs by skills type and level is essential.
- By combining various skills anticipation approaches, labour market actors can gain insight into potential skills requirements and imbalances, enabling informed decision-making.
- Although quantitative analysis provides partial information, qualitative methods are necessary to validate skills demand projections.
- Recognizing the effectiveness of skills anticipation systems involves combining multiple methods and addressing challenges beyond technical aspects, extending to institutional mechanisms. Quality data generation is crucial, but establishing and sustaining efficient and effective processes are equally vital.

Objective of skills anticipation[16]

In many employment impact assessment methods, the primary outcome variable is the number of employed persons. Employment impact assessments can extend their scope by incorporating skills requirements, provided that occupational data are available. Specifically, this approach, which involves a set of methods, can determine the workforce needed by skills type and level. Understanding the gap between future occupation and skills requirements and the current skills availability is crucial, both quantitatively and qualitatively.

Quantitative analysis of skills requirements typically relies on such indicators as occupational employment (based on the International Standard Classification of Occupations), educational attainment and qualifications as proxy measures for skills. These measures provide only partial information on skills that are important in the workplace. Due to significant gaps in data availability, qualitative analysis

16 Bolormaa Tumurduchur authored this section of the reference guide.

and methods are necessary for validating the quantitative modelling of skills demand. Various skills anticipation approaches can be combined to provide labour market actors with information on the potential future skills needs and imbalances, thus informing their decision-making.

An effective system for anticipating skills demand usually combines multiple methods and addresses challenges that extent beyond technical aspects to institutional considerations. For instance, while generating quality data is crucial, establishing efficient processes is equally vital.

▶ How the methods work

Delphi method: Delphi is a widely used exploratory technique aimed at structuring group communication processes to deal with complex issues. It is well-suited as a method for consensus-building by using a series of learning rounds to collect data from a panel of experts. The important characteristic of this method is that in the second and later rounds of the survey, the results of the previous round are given as feedback. The feedback process allows and encourages experts to reassess their initial judgements about the information provided in previous rounds. The Delphi method is mainly used to assess long-term issues (up to 30 years). For example, it is useful for predicting and assessing developments where there is no empirical database and where external factors are likely to have a determining effect.

Employer skills surveys: This is an instrument designed to generate data on employer demand for and investment in skills. It helps to define the type, level and composition of skills that individuals need to perform the work demanded by enterprises. This type of survey not only documents the skill content of current occupations but is also an appropriate tool for investigating future needs. It can be designed to obtain past or current information that may be used to anticipate which type of occupations will be in higher (or lower) demand or which skills will be key in future workers' skills portfolios. An employer skills survey usually collects information on employment within firms and requirements in terms of qualifications, skills and or tasks, along with information related to the firm itself or the sector.

Focus groups: A focus group is a form of qualitative research in which several people are asked about their perceptions, opinions, beliefs and attitudes regarding issues of interest. It supports the generation of ideas and encourages group thinking. The focus group concentrates on improving existing ideas. Preliminary ideas, products and versions of strategies are usually presented as a starting point. Participants are encouraged to express their own opinions and to interact with other members and the facilitator. Standard focus groups are more oriented towards obtaining information for researchers. Participants may include representatives of expert and technical staff involved in various aspects of skills anticipation and users of skills anticipation products.

Graduate surveys or tracer studies: These are surveys of graduates from education and training institutions. They are the most valuable single type of survey for collecting systematic and reliable information on the links between study and subsequent employment and work. Graduate surveys provide information on the whereabouts of graduates sometime after the awarding of a degree and allow this information to be linked with socio-biographic and study descriptors (such as gender, age, field of study, institution awarding the degree). These surveys provide valuable data for evaluating the results of the education and training of a specific institution, which may then be used for further development of the institution in the context of quality assurance.

Quantitative models: Such models help to develop an understanding of the economic reality, in particular trends and interactions within the economic indicators. These can be used under the assumption of relatively unchanged patterns to produce projections and forecasts of future trends or to simulate responses of the economy on different policy measures. Quantitative models are typically built using different statistical and econometric techniques. The development of such models is highly

dependent on the quality of input data, which are largely gathered by official authorities. Quantitative models for skills anticipation and forecasts require quite sophisticated information on occupational structure within sectors and/or qualification (education) structure within occupations. In practice, this process involves combining data that are usually derived from national accounts and surveys of households or the labour force.

Scenario development and foresights: Scenario planning is one of the best-known and most often cited techniques for preparing for the future. In foresight projects, the scenario method is a policy analysis tool that helps to describe a possible set of future conditions. At national, regional and local levels, scenarios can be used to improve planning capacity, enrich strategic public policy decisions and guide major capital investments. Scenarios are a preparation for potential future challenges – not predictions of what will happen. They help to identify future option spaces and enable confidence to act in a world of uncertainty. The method creates plausible views of the future that decision-makers can use to determine their best response and how to react to alternative situations. Scenarios are qualitatively distinct visions, told as stories, of what the future may look like. They make explicit the assumptions of how the world works.

Sectoral approach: A sectoral approach to skills anticipation is defined as one that looks at changing skills needs from the perspective of a particular sector. Such an approach includes any study, larger project or programme that adopts a methodological and analytical viewpoint of the sector. It also includes work carried out within a framework of institutional and stakeholder involvement (sectoral bodies, such as sectoral skills councils). In many respects, it makes more sense to talk about a sectoral focus rather than a sectoral approach or methodology. The sectoral approaches to anticipation and matching of skills are often included in the broader framework of the development of a sector in a country or region. The International Labour Organization's Skills for Trade and Economic Diversification is a sectoral approach designed to assist Member States in understanding strategic needs and challenges facing each target sector and identifying the skills development strategies required to address those challenges for future success. Combining evidence-based strategic analysis with stakeholder consultations, Skills for Trade and Economic Diversification emphasizes a participatory process that involves relevant ministries and their agencies, industry, social partners and education and training providers. This sectoral approach aims to deepen stakeholders' collaboration, build up capacity among national and sector partners and effectively move from diagnosing skill needs and challenges to implementing the required skills development responses, both at systemic and occupation-specific levels.

Big or real-time data methods: The use of big data analytics or of (almost) real-time information derived primarily from internet-based or digitized data sources (such as online job vacancies, users' CVs and financial transactions) is a rapidly growing area of business, research and policy domains. Rapid developments in the field of machine learning and algorithms (such as text mining and web crawling) have enabled the collection, synthesis and analysis of a rich and relatively unbounded set of data on skills and skills needs that were previously difficult or costly to collect using conventional sources, such as household or enterprise surveys. Innovative approaches include the exploration of big data analytics (for example, analysis of Google search keywords or online surveys on wages) and of data on vacancies collected by online job advertisement portals (such as the European Centre for the Development of Vocational Training's Skills OVATE[17] for gathering and analysing online vacancies in the European Union and the Flemish Public Employment Service uses text analytics on online job vacancies and CVs to match jobs and skills).

17 See www.cedefop.europa.eu/en/tools/skills-online-vacancies.

▶ Typical questions these methods can answer

- What are the current and future occupations in demand?
- What are the skills implications of a country's strategic priorities (growth potential, change in technology)?
- Which skills should a country develop towards the future (medium term or long term) and how can these skills be generated by the education and training system?
- What education levels and fields are most suitable for key occupations in a sector?
- What are the emerging occupations in the labour market?
- What demand for skills is likely to be newly emerging?
- How do the skills profiles of occupations change?
- What are the prospective fields of retraining and reskilling?
- What are the critical existing and future skills gaps?
- How can existing and future skills gaps be closed?
- How can skills demand be better anticipated in the future?

▶ Centrality of employment in the methods

Skills anticipation methods and tools are used to address a range of issues that countries face in developing skills for which there is a demand in the labour market. They also help to avoid skills mismatches that contribute to unemployment. Skills anticipation aims to provide information to all labour market actors about potential future skills needs and imbalances so that they can make decisions, develop measures and take actions to create better and more employment. Skills anticipation enables labour market actors to make better educational and training choices, and through institutional mechanisms, it leads to improved use of skills and human capital development.

▶ Pros and cons

Pros

- Holistic insights: Many methods, such as sector studies and foresight, offer detailed, comprehensive views within a specific context or sector.
- Direct user involvement: Methods, such as focus groups, surveys and vacancy studies, involve stakeholders directly, thus enhancing relevance.
- Specific and timely information: Big data and vacancy surveys capture timely, granular insights that are valuable for decision-making.
- Objective measurement: Quantitative forecasting and factual surveys aim for consistency and objectivity in data collection and analysis.
- Low-cost options: Graduate surveys, for instance, are relatively easy to execute and cost-effective compared with complex models.

Cons

- Subjectivity and non-representativeness: Qualitative methods and big data can introduce bias, lack generalizability and may not represent the full picture.
- High data and resource demands: Quantitative models and big data require substantial data inputs and technical capacity and can be costly to maintain.
- Potential inconsistency: Some methods, such as qualitative workshops and sectoral studies, may vary across sectors or lack systematic rigour.
- Sampling and response issues: Surveys often need a large sample and experience a low response rate, which can compromise accuracy.
- Short-term or partial coverage: Many methods focus on specific sectors, time frame or skill sets, thus limiting broader applicability.

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Experiment (randomized controlled trial)

Reference guide on employment impact assessment

Main points

- An experiment, specifically a randomized controlled trial, uses randomization to assign participants to treatment or control groups. It aims to identify causal effects of policies or interventions on specific outcomes and assess if observed changes are due to the intervention.
- A randomized controlled trial involves random assignments, like lotteries. Variations include a randomized phase-in design (staggered intervention start times) and a randomized promotion design (encouraging participation). These methods help evaluate interventions while addressing ethical concerns.
- Randomized controlled trials are increasingly used in employment studies, although their findings often have limited generalizability. Issues like low external validity and biases (publication and political) limit broader application.
- Advantages of the randomized controlled trials include high internal validity and simplicity in analysis. Limitations involve high costs, planning requirements and potential ethical concerns due to exclusion of control groups from interventions.

Objective of an experiment[18]

An "experiment" is a scientific approach that involves the use of randomization to define treatment and control groups. It is a type of empirical research that seeks to identify causal inferences to determine the effects of policies or interventions on outcomes of interest (Angrist and Krueger 1999).

The main method used in the experiment approach is the randomized controlled trial, in which people are allocated at random (by chance alone) to receive a treatment, such as participating in a specific intervention. The approach is extensively used in the field of labour economics and increased in

18 Drew Gardiner authored this section of the reference guide.

importance when one of the pioneers of the research method, the Jameel Latif Poverty Action Lab, was awarded the Nobel Prize in 2019.

The objective as an employment impact method is to determine whether the desired outcomes occurred and if those outcomes occurred because the policy was implemented. In other words, randomized controlled trials aim to determine whether observed changes in the economic or social well-being of beneficiaries can be attributed to a particular intervention, programme or policy. This is achieved by constructing a counterfactual scenario, such as aiming to answer the question, "What would have happened to the programme participant in the absence of the programme?"

▶ How the method works

The main application of a randomized controlled trial is the lottery. In a lottery, a sample of eligible individuals is randomly assigned to the two groups. Impact is the difference in outcomes between treatment and control. There are different ways of carrying out the randomization.

Variations

Randomized phase-in design

In this design, eligible participants are randomly assigned to different groups that receive the intervention at different times. Initially, one group receives the intervention while the other groups serve as controls. Over time, subsequent groups receive the intervention sequentially. This method allows all participants to eventually benefit from the intervention while still enabling the evaluation of its impact by comparing early and later recipients. This design helps mitigate ethical concerns related to withholding interventions. An example of this is the Galpão Programme in Brazil, which set out to improve the employment and earnings of at-risk youth in Rio de Janeiro's *favelas*. The programme provided vocational, academic and socio-emotional training over six months. Due to limited capacity, eligible participants were randomly assigned to start the programme at different times. This phase-in design allowed for a staggered implementation, whereby the outcomes of early participants were compared with those still on the waiting list, providing robust data on the programme's effectiveness (Calero and Rozo 2016).

Randomized promotion design

The randomized promotion design, also known as encouragement design, is a variation of a randomized controlled trial used when it is not feasible to exclude potential beneficiaries from an intervention. Instead of randomly assigning individuals to receive the intervention or not, this design randomly promotes or encourages the intervention among certain individuals. This encouragement could take various forms, such as information sessions, incentives or targeted communication efforts. The impact is then measured by comparing outcomes between those who were encouraged to participate and those who were not, using the encouragement as an instrumental variable to estimate the intervention's effect. An example of this is from Egypt, where researchers assessed the impact of a TV reality show on viewers' attitudes, employment and skills. The researchers implemented a non-symmetric encouragement design, which involved promoting the programme to a randomly selected subset of the target population. Participants received weekly text messages reminding them to watch the show and engage with its content (Gardiner et al. 2022).

▶ Typical questions this method can answer

- What is the overall intervention impact on outcomes A, B and C in group X and in context Y?

  For example, the average impact of the training intervention on the income of youth is an additional $20 per month. The intervention has a positive impact on participants' income.

- Do the outcomes vary across population groups?

  For example, the average increase in income is $40 for boys and nothing for girls. Older youth benefit more than younger youth (at $30 versus $10, on average). Therefore, the intervention is not equally effective for all participants.

- What is the short-term versus the long-term impact of the programme?

  For example, at the end of the programme, the average monthly income for participants is a negative $5 (a loss) compared with the control groups. Two years after the programme, the average increase in monthly income for the treatment group is $20.

- Is intervention design A or intervention design B more effective in terms of its impact on the direct beneficiaries?

▶ Centrality of employment in the method

The application of an experiment approach with employment assessments has grown considerably. A recent systematic review on youth-focused labour market interventions counted 220 experiments since 1990 (ILO 2022). Despite the success of this rigorous approach, including methodological innovations (multi-treatment arms, big data), the uptake among policymakers and social partners in labour institutions remains limited (Escudero 2022). A reason for this is the low external validity of these studies: The findings are restricted to narrow and limited circumstances and contexts, making the lessons difficult to generalize at the national, regional or global level. There has been evidence of publication and political bias, which cast doubt on the credibility of results (Miguel 2021; Bédécarrats, Guérin and Roubaud 2019). It is important to understand which policies actually "work" and are efficient - generating gains for participants that are large enough to justify their costs - and which ones are less successful.

▶ Pros and cons

Pros

- A randomized controlled trial performs well in terms of internal validity when appropriately applied. Lottery design is a robust method for developing a counterfactual because randomization implies no systematic differences between control and treatment groups (relying on fewer assumptions than other methods). It is therefore considered the most credible design to measure impact from the standpoint of individual outcomes.
- A randomized controlled trial is the simplest of all evaluation methods in analytical terms. The impact of the intervention in a random trial is simply the mean difference in outcomes between treatment and comparison groups.
- A randomized experiment allows for communities to be directly involved in the selection process for a fair and transparent allocation of benefits.
- It is easy to implement and communicate to programme staff.

Cons

- Conducting a randomized experiment can be cost- and time-intensive.
- No ex post implementation of this method is possible. Planning the evaluation must be part of planning the intervention (which is good practice in any case but does not always represent the reality in project work).
- A randomized trial requires a comparison group to be excluded from the intervention for the duration of the impact evaluation. Political and/or ethical concerns may emerge despite the transparent allocation criterion of randomization.

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Quasi-experiments

Reference guide on employment impact assessment

Main points

- Quasi-experiments lead to a better understanding of the impact of specific interventions on the post-programme outcomes of participants.
- The method is applicable to a broader set of interventions and can assess a broader range of outcomes than experiment studies.
- The method allows for the generation of evidence on what works and on why "what works" works.
- The method is relatively cheap and straightforward to implement ex post, in the presence of sufficient information on a comparison, or control, group, although it requires a degree of statistical knowledge and sophistication among the evaluators.

Objective of quasi-experiments[19]

The primary purpose of the quasi-experiment approach to impact evaluation is to determine the causal impact of a programme or intervention on the post-programme experiences of participants. The post-programme experiences of participants are compared with a control group of persons who did not participate in that programme.

How this method works

Quasi-experiment impact evaluation refers to a range of approaches that seek – through empirical and statistical means – to resolve the basic conundrum underlying impact evaluation. To estimate impact,

19 Niall O'Higgins authored this section of the reference guide.

"what happened" needs to be observed. But also, "what would have happened" in the absence of the programme or intervention needs to be determined by estimation. The unobserved (and unobservable) counterfactuals need to be estimated, typically by examining (and adjusting) the experiences of an appropriately identified control group.

The defining difference between the experiment and quasi-experiment approaches concerns the construction of the control group. With the experiment methods, the random allocation to the programme – or treatment – seeks to ensure that the group of non-participants – or control – does not differ from the group of programme participants in any systematic way that may affect post-programme outcomes independently of the treatment.

Quasi-experiment methods employ statistical corrections and/or programme design features to identify an appropriate control group ex post. These methods include:

Matching methods: Propensity score matching and other statistical corrections for sample selection bias seek to nullify any systematic differences between non-participants and participants so that any important factors that may affect differences in the post-programme outcomes experienced by the control group and by programme participants is factored out.

Difference-in-difference: This method first compares post-programme experiences with pre-programme experiences of participants. Then it does the same for a "similar" group of non-participants. Programme impact is then evaluated in terms of the difference (between control and treatment) in the difference (in pre-versus post-programme experiences) between the two groups in terms of the outcome of interest.

Regression discontinuity: This approach uses design features of the programme to establish a control group. The approach compares the (post-programme) experiences between participants and non-participants who are close to each other in some feature that determined their programme participation. For example, if a programme had an upper age limit of, say, 27, the experiences of 27-year-olds can be compared with a control group of 28-year-olds not eligible for programme participation.

Often, two or more of these methods can be combined to better identify the control group.

▶ Typical questions this method can answer

These questions are very similar to the questions that can be answered using the experiment approaches to impact evaluation, although, in practice, the quasi-experiment methods can be used to evaluate a broader range of outcomes for a larger group of interventions.

- What was the overall intervention impact on outcomes A, B and C in group X and in context Y?

  For example, the average impact of the training intervention on the income of youth was an additional $20 per month. The intervention had a positive impact on participants' income.

- Did the outcomes vary across population groups?

  For example, the average increase in income was $40 for boys and nothing for girls. Older youths benefited more than younger youths (at $30 versus $10, on average). Therefore, the intervention was not equally effective for all participants.

- What was the short-term versus the long-term impact of the programme?

  For example, at the end of the programme, average monthly income for participants had decreased by $5 (a loss) when compared with the control group. Two years after the programme, the average increase in monthly income for the treatment group was $20.

- Did the intervention have spillover effects?

  For example, not only did participants have a $20 higher average income but their neighbours also experienced a $5 increase.

- Was intervention design A or intervention design B more effective?

Centrality of employment in the method

Impact evaluations using either experiments or quasi-experiments have traditionally focused on employment and income benefits for participants in active labour market programmes, which include four broad classes of interventions – (i) training programmes, (ii) subsidized employment programmes, (iii) job search assistance programmes and (iv) support for entrepreneurship. In recent years, quasi-experiment-based analyses have broadened the types of programmes analysed, and both the experiment and quasi-experiment analyses have widened the range of outcomes considered. For example, there have been several impact evaluations of programmes seeking to reduce informality (see, for example, Jessen and Kluve 2021) and quasi-experiment studies examining the impact of the interactions between, for example, active labour market interventions and other labour market institutions, like employment protection legislation (see O'Higgins and Pica 2020). Studies now also look at outcomes, such as the impact of interventions on participants' mental health (Wang et al. 2021). These evaluations are allowing a better understanding not only of what works in labour market policies and interventions but also, crucially, why «what works" works.

Pros and cons

Pros

- The quasi-experiment method is much cheaper than the experiment method.
- Such methods are also applicable for determining a broader range of outcomes in a variety of contexts than experiment evaluations.
- Some of these methods can consider the effects of programmes on the broader environment, such as in the application of difference-in-difference approaches to large-scale interventions likely to also impact the labour market experiences of non-participants.
- They do not typically require ex ante action at the programme level, in contrast to the experiment method, although, depending on the specific method applied, pre-programme planning of control group identification may be advisable.

Cons

- Appropriate implementation of the methods requires much greater knowledge and understanding of quite sophisticated statistical techniques and hence must be implemented by adequately trained staff.

- These methods employ a variety of techniques to appropriate identify an "as if" randomly chosen control group. Yet, they are not randomly chosen, and verification of the appropriate identification of the control group can be challenging.
- External validity is an issue also for this approach.

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Meta-analysis

Reference guide on employment impact assessment

Main points

- Meta-analysis provides reliable estimates of relationships between phenomena, based on multiple studies.
- The method requires studies under analysis to be sufficiently similar to account for differences quantitatively or qualitatively.
- It involves several steps, such as determining research questions, sampling relevant studies, setting inclusion criteria and coding data.
- Meta-analysis helps to generalize results and improve parameter estimates of the effects of interest.

Objective of meta-analysis[20]

The purpose of meta-analysis is to provide reliable estimates of relationships between one phenomenon and another, based on a range of studies. An example is the impact of training programmes on the post-programme employment prospects of participants. Where sufficient estimates are available, meta-analysis can further seek to identify the causal factors underlying the relationship – not just "what works" but "why what works works". Meta-analyses can be performed where there are multiple studies seeking to determine the same or similar effects. It is, in essence, "a quantitative survey of a literature reporting estimates of the same parameter" (Paldam 2015). Studies under analysis must be sufficiently similar so that their differences can be accounted for in an appropriate quantitative (or qualitative) manner.

20 Niall O'Higgins and Juan Chacaltana authored this section of the reference guide.

How this method works

Meta-analysis is a statistical approach that synthesizes the results of multiple studies to generate more comprehensive and reliable conclusions than those drawn from individual analyses. This process begins with the identification of relevant studies through systematic searches of academic databases, ensuring that studies meet established criteria. Next, the quality of each selected study is assessed to gauge the reliability of their findings. Specific data, such as effect size and sample size, are then extracted. The meta-analytic technique typically involves calculating a weighted average of effect sizes, giving greater weight to studies with larger samples or higher quality. An essential step in this process is assessing heterogeneity, which evaluates whether differences between studies could reflect varying populations or contexts. The final interpretation of results offers insights into the overall strength and direction of relationships between the studied variables. Meta-analyses increase statistical power, enhance precision and resolve conflicts across studies while also revealing underlying patterns. However, their validity is dependent on the quality of the included studies and may be influenced by publication bias or substantial heterogeneity, which can limit the generalizability of the findings.

Typical questions this method can answer

Meta-analysis can be used to answer two fundamental questions in relation to estimates in existing studies (Paldam 2015):

- Do the estimates converge to an average that provides a "good" approximation of the true relationship?
- Can we identify the factors underlying variations in the estimated effects?

Paldam (2015) argued that meta-analysis is better at answering the first of these questions. In practice, meta-analyses have been performed in a variety of fields. For the labour market, they have been used to assess the effects of minimum wage on employment (see, for example, Card and Krueger 2007) or on the effects of active labour market policies (see Card et al. 2017). Performing a meta-analysis involves several steps:

- Identification of a research question and a justification for the meta-analysis.
- The sampling process involves determining relevant impact evaluations or quantitative assessment studies. It is typically based on a search of multiple sources and databases. First, target websites and databases or other methods to search for information (such as networks) need to be defined. Second, strings or specific titles need to be defined, which will be used to systematically identify relevant content in previously defined sources.
- Inclusion criteria of the potential lists include defining the languages in which the search will be conducted and what type of studies will be covered. Typically, the selection of studies includes an assessment of the quality of those articles, based on, for example, whether studies eliminated biases in their analyses, as well as risks of self-selection. To achieve this, studies can be selected based on their methodology.
- The analysis includes a process of coding and classification of the estimates, including intervention characteristics, target groups, period and scope of the interventions, outcomes (typically income and employment but also outcomes related to the quality of employment), unit of observation and characteristics of the sample and techniques used in the studies.

Once the data set of estimates, characteristics and covariates is completed, meta-analysis is used to summarize the results and to gain an understanding of whether the effects can be generalized. The goal is to derive improved parameter estimates of the effects of interest.

▶ Centrality of employment in the method

With a focus on employment outcomes, the ILO recently promoted studies that use this method.

O'Higgins and Moscariello (2017), for instance, employed meta-analysis to look at the impact of labour market institutions on youth labour market outcomes. Specifically, their paper examines the effects of minimum wages on youth employment, based on 328 estimates from 15 countries. The meta-analysis first confirmed the findings of the literature review reported in O'Higgins (2001), mainly that, in the overwhelming majority of cases, the estimated negative effects of minimum wages on youth employment was either zero (not statistically significant) or very small. Hence, removing or lowering minimum wages for young people is unlikely to be an effective tool for improving the employment prospects of young workers. The meta-analysis further identified the presence of strong complementarities between minimum wages and other labour market institutions. For example, minimum wages and employment protection legislation are mutually supportive institutions, so that any potential negative youth employment effects associated with the introduction or raising of minimum wages will be attenuated in the presence of strong employment protection legislation. In high-income countries, minimum wages have smaller dis-employment effects in the presence of the appropriate collective bargaining arrangements – specifically in the presence of strong worker representation, accompanied by coordinated but decentralized collective bargaining arrangements. Overall, minimum wages tend to have more detrimental effects on youth labour markets in low-income countries where, inter alia, labour market institutions are weaker. This thus suggests that minimum wages are a more useful instrument when other protective labour market institutions are already in place and operating effectively.

Study	Methodology	Focus	Primary finding
O'Higgins and Moscariello (2017)	Meta-analysis: 328 estimates from 15 countries	Impact of minimum wages on youth employment	Minimum wages have little to no negative impact on youth employment, especially in countries with strong labour market institutions.

Kluve et al. (2017) analysed the effects of different interventions on labour market outcomes of youths. They further explored the role of context and programme design. They conducted a systematic review of empirical evidence on the labour market outcomes of active labour market programmes targeting youths worldwide. Systematic reviews are a form of meta-analysis that satisfy a series of additional requirements, for example, regarding the criteria for the inclusion of studies and estimates. Interventions in the Kluve et al. review included skills training, entrepreneurship promotion, employment services and subsidized employment. Outcomes of interest included employment, earnings and business performance. The emphasis of the meta-analysis was on studies conducted in low-, middle- or high-income countries. A systematic search was performed for relevant evidence across more than 70 sources, using search terms in English, French, German, Portuguese and Spanish. After reviewing some 30,000 registries, 113 eligible impact evaluations were identified, including 2,259 effect sizes. The meta-analysis indicated positive effects of entrepreneurship promotion and skills training on employment and earnings. Effects of employment services and subsidized employment were generally small and

not statistically significant. The analysis revealed larger programme effects in low- and middle-income countries than in high-income countries and in programmes targeting disadvantaged youths. In general, comprehensive programmes comprising multiple elements appeared to be more effective.

Study	Methodology	Focus	Primary finding
Kluve et al. (2017)	Systematic review: 113 eligible impact evaluations, including 2,259 effect sizes	Effectiveness of active labour market programmes for youths	Entrepreneurship promotion and skills training are effective in improving youth employment and earnings, especially in low- and middle-income countries.

More recently, a joint ILO and World Bank study (Puerto et al. 2024) examined the impact of youth-targeted active labour market programmes on employment and earnings, based on a systematic review of evaluations conducted from 1990 to 2021. The evidence showed these interventions significantly enhance employment and earnings outcomes for young people, particularly in low- and middle-income countries. The study, encompassing 228 impact evaluations of 220 interventions across 62 countries, revealed that programmes targeting marginalized youths, such as women and low-skilled individuals, exhibited the most substantial benefits. Notably, entrepreneurship support and training programmes had the largest global impact, with specific effectiveness observed in employment services in low- and middle-income countries and skills training in high-income countries. The analysis identified several critical design features of successful programmes, including the integration of soft skills training, certification and multifaceted approaches to address various barriers to employment. The collaboration between public and non-public actors and the localization of programmes also contributed to enhanced outcomes. Despite the positive findings, the summarizing paper underscores the need for further rigorous impact evaluations and highlights the substantial variation in programme costs.

Study	Methodology	Focus	Primary finding
Puerto et al. (2024)	Systematic review: 228 impact evaluations of 220 interventions across 62 countries	Impact of youth-targeted active labour market programmes	Youth-targeted programmes, especially those targeting marginalized groups, can significantly improve employment and earnings outcomes. Entrepreneurship support and training are particularly effective.

Escudero et al. (2019) compiled a meta database of 51 studies on active labour market policies in general (for youths and adults), highlighting the Latin American and Caribbean experience. They extracted a sample of 296 impact estimates. The analysis included interventions in skills training programmes (around 75 per cent of the estimates) and labour market intermediation services, private sector incentives (such as employment subsidies and self-employment and microenterprise creation programmes) and public works. Outcome variables included earnings and employment and some measures of job quality, such as hours worked and formality rates. The study found that these active labour market programmes in Latin America and the Caribbean were particularly effective in increasing the probability of having a formal job, compared with other outcomes and, in particular, that the training programmes were slightly more effective than other types of intervention. In fact, formal employment was also the outcome category that was most likely to be impacted positively by these programmes. In terms of targeting, the study found that active labour market programmes in Latin America and the Caribbean worked better for women than for men and for youths when compared with prime-age workers. Additionally, the medium-run estimates were not more likely to be positive than short-run estimates, while programmes

of short duration (four months or shorter) were significantly less likely to produce positive effects when compared with more prolonged interventions.

Study	Methodology	Focus	Primary finding
Escudero et al. (2019)	Meta-analysis: 51 studies, 296 impact estimates	Effectiveness of active labour market policies in Latin America and the Caribbean	Active labour market programmes in Latin America and the Caribbean are effective in increasing formal employment, especially for women and youth.

Jessen and Kluve (2019) explored the effectiveness of formalization interventions in low- and middle-income countries. Interestingly, although a multitude of interventions have been implemented to increase the formalization of workers and economic units, there is little empirical evidence on these initiatives. The study included the following types of intervention: information campaigns, simplified registration procedures, reductions of payroll taxes and interventions enforcing formalization. They identified 32 academic studies and compiled a database of 157 impact estimates that empirically evaluated one or more of these formalization interventions. The quantitative analysis correlated the impact estimates of the studies – effect size, sign and statistical significance – with explanatory factors, such as intervention type, outcome variable, scope of the intervention and other covariates. Several important findings emerged: First, the intervention type was not a strong determinant for the effectiveness of formalization interventions. Second, the outcome "worker registration" showed significantly better results than other outcomes (for example, firm registration). Third, interventions at scale were more effective on average than singular programmes. This was in line with International Labour Organization Recommendation No. 204 concerning the Transition from the Informal to the Formal Economy, which requires an integrated approach to facilitate that transition.

Study	Methodology	Focus	Primary finding
Jessen and Kluve (2019)	Systematic review: 32 academic studies, 157 impact estimates	Effectiveness of formalization interventions	Formalization interventions can be effective, but their effectiveness depends on factors like intervention type, outcome variable and scale.

▶ Pros and cons

Pros

- Meta-analysis directly addresses the issue of external validity – a weak point in individual studies of impact.
- Given the inclusion of a sufficient number of independent estimates in the meta-analysis, the method can provide additional information on the source of variations in intervention impact: for example, why the effects of minimum wages on employment are negative in some circumstances and positive in others.

- Meta-analysis thus seeks to identify both what works and why it works. It provides a basis for better-designed programmes applicable in different contexts and that go beyond the findings of specific impact evaluations.

Cons

- The method is susceptible to publication bias.[21] There are several tests that can be employed to reduce this problem. The primary issue is the quality of the underlying estimates. Systematic reviews seek to address these issues by specifying a series of additional criteria that must be fulfilled for the inclusion of studies in meta-analysis.

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21 Publication bias arises when studies are more likely to reach the light of day if they obtain one type of a result rather than another.

Geographic information system and remote sensing

Reference guide on employment impact assessment

Main points

- A geographic information system and remote sensing are powerful tools for understanding the relationships, patterns and trends in geographic data, aiding in economic and employment impact analysis.
- Combining administrative and household survey data with geographic information enhances the assessment of changes in employment outcomes due to policy interventions.
- Night-light data serve as a proxy for gross domestic product, allowing the monitoring of economic activities and labour market changes with high temporal frequency in data-scarce regions.
- Remote sensing is invaluable for monitoring local economic growth, policy interventions at the meso level and project level, environmental changes (such as drought) or large shocks (such as natural disasters and conflicts) as well as monitoring their employment outcomes, especially in medium- and low-income countries.

Objective of a geographic information system and remote sensing[22]

A geographic information system, commonly known as GIS, is a powerful tool that integrates and analyses spatial or geographic data. It enables users to visualize, question, analyse and interpret data to understand relationships, patterns and trends in a geographical context. With a geographic information system, the main dimension of analyses is the location variable. The system covers a large range of approaches and data because it encompasses any data that contain location, such as longitude and latitude, and time dimensions.

22 Alina Game and Matthieu Charpe authored this section of the reference guide.

This reference guide covers only certain applications of a geographic information system related to the measurement of economic and employment impacts. A first set of applications covers traditional administrative data, such as population census, labour force surveys or household income and expenditure surveys. These surveys are standard surveys to document the world of work and the labour market. A geographic information system can bring new insights in labour economics by studying the impact of location of economic agents on labour market outcomes, such as employment participation, unemployment, employment status or labour incomes.

This guide also features remote sensing, which covers data collected from above the Earth's surface by planes or satellites using sensors. In recent years, the mobilization of remote sensing in economics has contributed towards overcoming limitations related to the lack of data in medium- and low-income countries. The appealing aspect of remote sensing is that it is often publicly available data with a high time frequency (daily, weekly or monthly data), covering the entire Earth's surface, with a small geographic granularity (at 1 kilometre or less). It is measured by a machine, the sensor, which avoids many biases present with existing administrative data.

There has been growing interest surrounding the utilization of a geographic information system and remotely sensed data within economics. Satellite data have been applied to economic analysis across many fields to investigate changes in infrastructure, agricultural productivity, inequality and in the wake of natural disasters.

▶ How the methods work

Geographic information system

- Census, household surveys and labour force surveys collect administrative data to produce national statistics on employment and shed light on the structure of the labour force. Labour force surveys, usually collected from a sample of individuals that is nationally representative, aim to capture the number of jobs and the characteristics of these jobs. This information can be used to track progress and for monitoring and planning purposes.

- Alongside employment data, geographic information is also obtained during these surveys, usually through the collection of Global Positioning System, or GPS, coordinates of surveyed households or location and the delineation of enumeration areas used in the survey. Due to the sensitivity of location information, the data are not made publicly available but can be accessed through collaboration with the relevant statistical office. Being able to analyse the location and distribution of employment related indicators can provide insight for planning and policy-setting.

- Combining the microdata with other geographic data, such as the location of infrastructure, is valuable for conducting an impact assessment to understand how employment outcomes have changed as a result of an intervention. If data are available from two time periods – before and after an intervention, statistical techniques, such as difference, can be applied to measure changes in employment outcomes over time for individuals impacted by an intervention. This could be applied to measure the impact from a range of programmes or events, such as the building of a new road, the occurrence of conflicts or natural disasters, or to measure the impacts of environmental changes, for example.

- A challenge associated with using this type of data for measuring employment impacts is based on the availability of data, particularly in low- and middle-income countries. Lack of up-to-date surveys or data collection at regular time intervals can make it difficult to have enough data from suitable

time periods to measure impacts. Access to sensitive information, including the location data that are necessary for analyses when examining impact at a local level, is understandably restricted and requires collaboration with the government and national statistical office. There is also a need for cleaning and processing the survey data to extract relevant indicators and ensure accuracy of the data before being used in analysis.

Remote sensing

- The use of satellite images in combination with algorithms and cloud-based solutions have had a deep impact in economics. The main advantages are the low cost of data collection, the repeated observation of the data and the large scale of data collection. Applications in economics are related to night lights, climate and weather, topography, agricultural land use and crop choice, urban land use, natural resources and pollution monitoring, among others (Donaldson and Storeygard 2016).
- This guide discusses night-light as well as land-use data. Night-light data are the amount of light at night that can be observed from outer space. Satellites initially designed to collect environmental information at night can detect light from human activities (Elvidge et al. 2007). Three publications have shown the ability of light at night to predict gross domestic product in countries with underfunded statistical systems: Henderson 2012; and Chen and Nordhaus 2011; Ghosh et al. 2010.
- The data are available since 1992 and are divided into two time periods (as a result of the change in sensors in the early 2010s, with improved radiometric properties). The US Air Force Defence Meteorological Satellite Program generated data for 1992–2013. The Visible Infrared Imaging Radiometer Suite is an instrument developed by the National Aeronautics and Space Administration (NASA) and the National Oceanic and Atmospheric Administration. It provides night-light data since 2012 at a daily, monthly and annual frequency.
- The main challenge with the use of night lights is to distinguish changes in light related to human activities from changes in lights related to non-human activities as well as changing atmospheric conditions. It follows that heavy data processing is needed to remove lunar lights, auroral activities, biomass burning, gas flare, cloud cover or snow albedos. Calibrated and corrected products are available online from the Earth Observation Group at the Colorado School of Mines as well as from the Blackmarble suite produced by NASA.[23]
- Night lights have been shown to be a powerful instrument to track economic activities in countries with limited GDP data. Night-light data can also be used to indirectly track the evolution of the labour market. The methodology commonly followed in development economics is to use the light intensity as a proxy for GDP and to aggregate pixel data at the geographic level desired (village, city, county, region or country). The use of data with a high time frequency allows for the application of both conventional panel regression methods and quasi-experiment techniques. In addition, the small geographic granularity in the data makes it possible to assume that a pixel is a representative economic agent, thus creating a bridge between macro, meso and micro approaches.

23 See Defence Meteorological Satellite Program data at https://payneinstitute.mines.edu/eog/ and VIIRS data at https://blackmarble.gsfc.nasa.gov/#home.

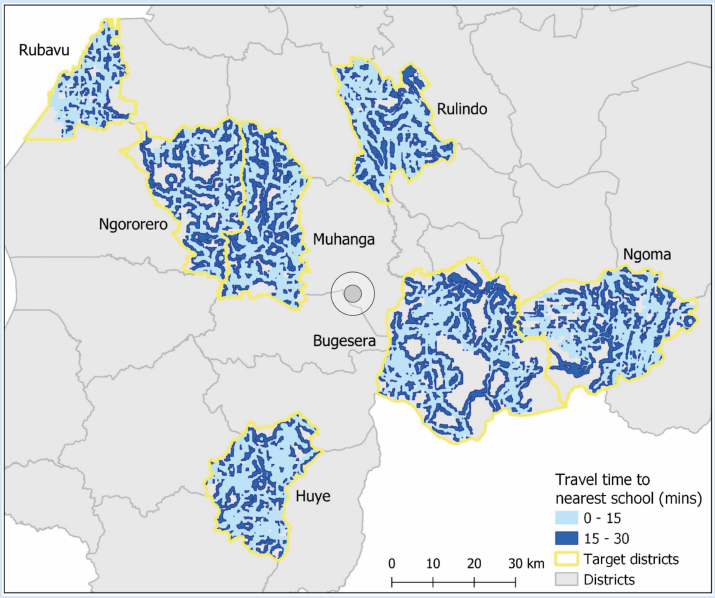
▶ Typical questions these methods can answer

Impacts of transport infrastructure

The construction or rehabilitation of transport infrastructure, particularly roads, can drive economic development and improve employment outcomes by reducing transport costs and increasing access to the labour market for a population. Capturing the long-term impacts of these interventions is often difficult, but a geographic information system and remote sensing can be used to answer a variety of questions, such as:

- What are the long-term economic and employment impacts of road interventions? See the employment impact assessment of a transport corridor in Kenya (Game and Kang 2023a) that used night-light data to estimate the impacts on GDP and employment over time. Lights were used as a proxy for GDP and employment to highlight an increase in economic activity due to a road intervention. To further examine employment effects, see also the employment impact assessment of a road programme in Rwanda (Game and Kang 2023b) that used labour force survey data to analyse changes in employment dimensions for feeder roads.

a. Figure 1 - Travel time to schools from improved road access in Rwanda (2023)



Source: Game and Kang, 2023b

- How have roads improved access to social infrastructure? See also the Rwanda study that assessed how rehabilitated roads improved access to health and education infrastructure for the population by using the geographic information system data to simulate travel time scenarios to health facilities and schools (Game and Kang 2023b).

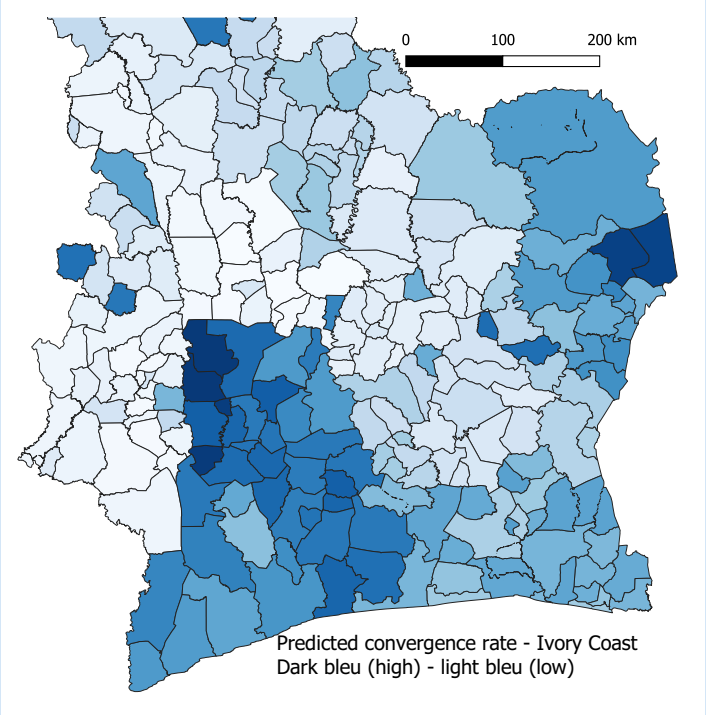
Productivity growth and structural transformation

Labour productivity is an important dimension of economic progress and development. Tracking the evolution of labour productivity is crucial for the evaluation of public policies. In low-income countries, national account data do not make it possible to measure labour productivity other than at annual frequency and country aggregation.

A geographic information system offers the possibility to build a proxy for labour productivity by combining night-light data and high-resolution population data at pixel level. For example:

- What is the trend in labour productivity at the subregional level in sub-Saharan Africa? And what are the factors shaping these trends? Charpe (2023) demonstrated the usefulness of this approach to track convergence in labour productivity between richer subregions and less-developed subregions in ten African countries, including Ivory Coast. This work showed that despite a non-negligeable convergence rate, the poorest subregions were left behind. Looking at the factors behind the (absence of) convergence, the study pointed to the importance of sector specialization, especially in manufacturing.

b. Figure 2 – Sub-regional labour productivity trends in Côte d'Ivoire (1998-2014)



Source: Charpe, 2023

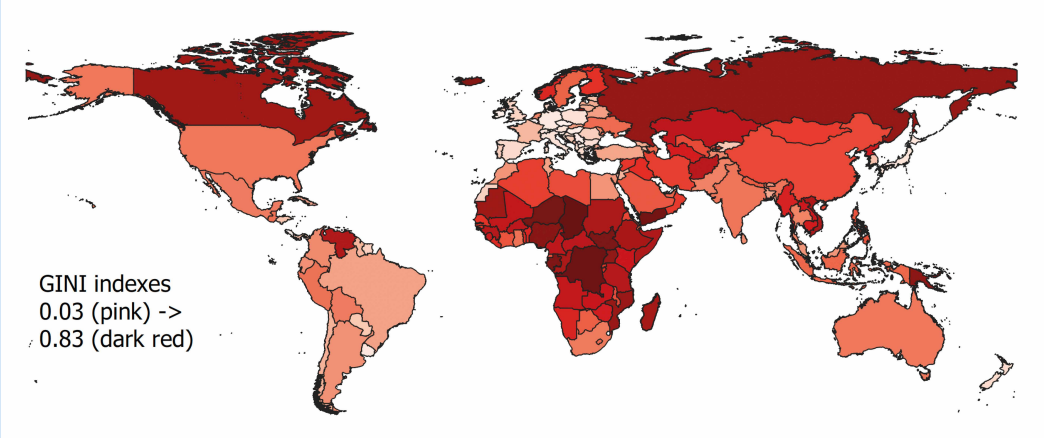
Inequality

The question of inequalities is an important element of the mandate of the International Labour Organization, as stated in its Constitution and in the Declaration of Philadelphia. The ILO adopted a strategy to reduce inequalities in 2022, and reducing inequalities is at the centre of the social justice coalition. Measuring inequalities is often challenging, especially in the poorest countries that lack well-funded statistical offices. A geographic information system can contribute to measuring and analysing inequalities where traditional data are missing. For example:

- What is the level and trends in inequalities? Spatial GINI indexes can be built from the geographic distribution of night lights. The exhaustive country coverage, especially regarding low-income countries, shows that spatial inequalities are higher than previously measured. The availability of data since 1992 can also inform policymakers about the dynamic of inequalities over time.

Regarding spacial inequalities, the decline in inequalities has only materialized in middle-income countries, while low-income countries have made little progress in terms of reductions. In high-income counries, although inequalities are relatively low, there has been no progress in reducing inequalities further in recent years (ILO, forthcoming).

c. Figure 3 – Night light as an indicator of global spatial inequalities (2023)



Source: ILO, 2025.

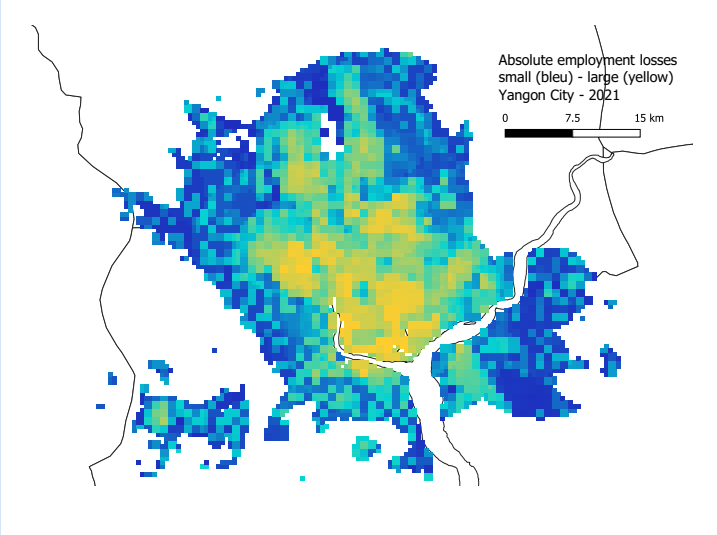
Conflicts and natural disasters

Emergency impact assessments in the aftermath of a conflict or natural disaster relies heavily on geographic information system techniques (O'Higgins 2025). Access to an impacted area is made difficult by the crisis, and gathering information through traditional surveys is often not possible. Remote sensing can help understand the severity of the crisis as well as the size of the population impacted and their needs for assistance (ILO 2024a, 2024b). For example:

- What is the population in the vicinity of conflicts? Combining geo-refrenced data on conflicts and conflict-related casualties with high-resolution maps of populations, it is possible to quantify the population in the proximity of conflicts globally. In the report GET YOUTH 2024, the ILO estimated that 320 million persons were in the vicinity of conflicts in 2022, including 56 million youths (ILO 2024b).

- What is the impact of conflicts on the labour market? Taking the example of the conflict in Myanmar in 2021, the use of night-light data contributed to quantifying the impact of conflicts on employment losses at pixel level. Of the 9.5 million workers located in the conflict areas in Myanmar, employment losses amounted to 1.1 million jobs in 2021 and 800,000 jobs in 2022.

d. Figure 3 – Impact of conflict on employment losses in Myanmar presented by night-light data (2021)



Source: ILO, 2024b.

Environment and agriculture

The impacts of climate change and environmental degradation are major challenges for development, including employment and livelihoods for people in many regions of the world. Satellite data can be used to measure indices related to the environment, such as vegetation productivity, soil moisture and land cover, which, when compared over time, can give an estimation of the severity of events and the subsequent impacts on agriculture and livelihoods. Satellite data can be further applied to identify crop types and to track and estimate yields for agricultural management and mitigation. For example:

- How has drought impacted crop productivity and employment outcomes? The ILO has undertaken a study on using satellite data to measure drought severity and the impact this had on main crop yields in Zambia (Dadhich and Game, forthcoming). Imagery was used to highlight drought areas and identify cropland. Microdata were then combined to examine the effects on employment outcomes before and after the drought event by measuring changes in employment, wages and working time for the affected population.

▶ Centrality of employment in this method

With the use of night-light data, satellite images measure electricity consumption, which is an indirect measure of economic activity. The data do not capture a direct labour market characteristic. But it is possible to translate changes in light into changes in GDP and employment based on country-level panel regressions. Many studies have shown that the correlation between GDP and light is stable and close to unity for the most recent data (Beyer, Hu and Yao 2022; Chen and Nordhaus 2019). The elasticity appears to be inversely related to the income level of a country.

In low-income countries, it is likely that an increase in GDP per capita generates more light intensity than in high-income countries. Additionally, the relationship is stronger in urban areas because an increase in agriculture production translates into light primarily through the income channel of farmers rather than direct light emissions.

There is a proposal to complement night-light data with land cover data to better capture agriculture GDP (Souknilanh, Andersson and Hall 2015). The relationship between lights and employment is also stable and consistent with light to GDP elasticity and Okun's coefficients. Based on country-level panel data, the light to employment elasticity is around 0.7.

▶ Pros and cons

Night lights

Pros

- Night-light data are collected at regular time intervals (monthly, annual and daily) and high spatial resolution, approximately 450x450 metres.
- The data are available as open source, with processed and clean products available.
- Night lights can be used as a proxy for GDP and employment.

Cons

- Further pre-processing of data is often required.
- The data source changed in 2013, and there is now a need for harmonization if using both data sets.
- The data can only be applied as a proxy – detailed employment information needs to be combined with microdata.

Applications

- Build a proxy for GDP and employment.
- Construct spatial GINI indexes to measure inequality.
- Measure labour productivity.

Satellite imagery

Pros

- Satellite imagery is collected at regular time intervals, with many different sources.
- It has high spatial resolution, and there is great variety available as open source.

Cons

- Raw satellite data require pre-processing to calculate indices.
- Higher resolution data are available but through commercial companies.
- The data need to be combined with microdata to be used to analyse employment.

Applications

- Construction of environmental indices – soil, vegetation, water.
- Change detection through comparison of images over time.
- Landcover classification.

Administrative and survey data (labour force survey, household budget survey, census)

Pros

- This type of data provides additional information related to employment and demographics, often processed and cleaned in-country.
- It is representative of the population, with geospatial information collected, such as with household coordinates.

Cons

- Geospatial information is usually sensitive and needs to be accessed through the national statistics office.
- It is not always collected at regular time intervals.
- Its availability and accessibility differ between countries.
- Data collection issues can lead to problems with data quality.

Applications

- Measuring employment impacts of different projects, events and interventions.
- Highlighting variations in employment outcomes at the administrative level.
- Analysing changes in employment over time (if data available across time periods).

Georeferenced data (location of infrastructure, roads, markets and location of events, conflicts)

Pros

- It is straightforward data, easy to obtain and apply.
- It is freely available and collected through various sources.
- The location of many different infrastructure or events are available that can easily be combined with additional layers to answer employment questions.

Cons

- Georeferenced data are not always regularly updated and require some quality assurance.
- There may be need to edit or digitize own data if available data are not suitable.

Applications

- Accessibility analysis of roads to social infrastructure.
- Locations of conflicts and impact on population.
- Planning and targeting of infrastructure based on existing coverage.

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Informal employment impact assessment

A reference guide on employment impact assessment methods

Main points

- Quantitative surveys can provide valuable information on the characteristics of informal businesses and workers and the workings of the labour market.
- Consecutively applied surveys permit the study of the evolution of the informal economy and, if they are based on panels, additional conclusions can be drawn on the factors that favour the transition to formality and the impact of formalization.
- The econometric analysis of survey data permits establishing a relationship between the characteristics of informal businesses and workers, the conditions for formalization and the dynamics of a labour market.
- Satellite accounts identify an informal sector's contribution to different components of national accounts and the evolution of informal employment, covering the entire informal economy.
- Regression and decomposition analysis can contribute to identification of macro-level factors that influence the relative size of informal employment.
- Instruments that generate qualitative information on attitudes, preferences and experiences of informal businessowners or workers (surveys, in-depth interviews, focus groups) contribute to the understanding of the heterogeneity of firm and worker informality.

Objective of informal employment impact[24]

For assessments of the impact of structural changes and specific policies on informal employment and the transition to formality, a range of instruments is used. The most relevant are general equilibrium models and, especially, experiments and quasi-experiments (Weller 2022). Meta-analysis has been used to summarize and assess the results of these exercises.

24 Jürgen Weller authored this reference guide.

Other instruments for analysing the characteristics and dynamics of informal employment and the role of factors that have relevant impact on them are quantitative and qualitative surveys, econometric techniques for the exploitation of the data they generate and other variables and surveys for the development of satellite accounts that enrich traditional national accounts, as well as decomposition exercises. Although these tools rarely are able to establish causal relationships between specific policies or structural trends and informal employment, they provide valuable information on the correlation between relevant variables, which is useful for understanding the workings of labour markets and for policymakers´ options for promoting the transition to formality.

▶ How the method works

When assessing the impact of informal employment, the main quantitative business surveys that provide information on the characteristics of informal businesses and workers are small and medium-sized enterprise surveys, specific informal business surveys and 1-2-3 surveys, which, in a first phase, take advantage of a household survey with which informal enterprises are identified and whose characteristics are surveyed in the next step (Nordman and Roubaud 2010). Consecutive surveys permit the analysis of the evolution of the informal economy and, if the survey is based on a panel, additional conclusions can be drawn on the factors that favour the transition to formality. In addition, household surveys generate valuable information on the characteristics of informal workers (self-employed and employees). The 1-2-3 surveys can be used as an additional input for the development of satellite accounts on the informal economy, which are an extension of the standard national accounts. In such case, the survey data are combined with data originating from national accounts on production accounts by activity, production segment and institutional sector. An employment matrix is also constructed by category of occupation, activity, production segment and institutional sector (INEI 2019).

Qualitative surveys aspire to identify attitudes, preferences and experiences of informal businessowners (small enterprise employers or self-employed workers) or, exceptionally, informal employees. In contrast with quantitative surveys, this type of survey "does not aim at establishing frequencies, means or other parameters" (Jansen 2010) but pretends to cover the heterogeneity of situations within the population of interest. For this reason, the qualitative survey sample does not need to be representative for this population, but it must aspire to cover the diversity of interests (businesses in different regions or industries).

These surveys may include open questions that allow for differentiated answers regarding complex issues. This type of question is used during in-depth interviews with micro-entrepreneurs (as in Selamat et al. 2011) or in focus groups (such as with experts on informality in ILO 2021).

Qualitative surveys may also cover limited quantitative information on the characteristics of businesses and individuals. Similarly, qualitative questions can be included in quantitative surveys (with representative samples) so that the diversity of the results of the first can be related to the characteristics of informal businesses identified by the latter (see Cling, Razafindrakoto and Roubaud 2012).

The data generated by both qualitative and quantitative surveys permit statistical presentation of findings. Additionally, the information generated through quantitative surveys can be analysed by econometric techniques, which permit insights on the characteristics and dynamics of the informal economy, for example, by a cross-section comparison or time series analysis.

Econometric techniques, especially regression analysis, also have been applied to macrodata to identify factors that determine the proportion of informal employment.

With decomposition methods, the evolution of the rate of employment formality can be divided analytically by the change of this rate on the industry level or in the formal and the informal sectors

(change within) and the change of the relative weight of industries or the formal and informal sectors in the employment structure (change between), which is interpreted as structural change as a result of differentiated economic growth rates (see Salazar-Xirinachs and Chacaltana 2018).

▶ Typical questions these methods can answer

Quantitative surveys on the informal economy permit a characterization of informal economic units, relating the formality or informality status to variables, such as size (by revenue or number of workers), firm age, composition of the workforce, industry, location and technology level, as well as personal characteristics of the owner or manager.

Household surveys permit the characterization of different types of informal workers and their comparison with formal workers, which can provide clues on obstacles to formalization on the supply-side of the labour market. If a survey has panel characteristics, it can be used for the analysis of labour market dynamics, such as workers´ transition between formal and informal jobs or other labour market statuses.

Satellite accounts expose the informal sector's contribution to different components of a country's national accounts and the evolution of informal employment (both within and outside the informal sector, thus covering the entire informal economy). If calculations are made on a regular basis, this information can be used in econometric exercises that establish relationships with factors that may explain changes in the informal sector's contribution to gross domestic product. It is also possible to model the consequences for informal employment of changes in the national accounts data, to which the satellite account is linked.

The most obvious motive for the application of qualitative surveys among the owners of informal or recently formalized businesses is to identify different issues related to formality or informality in the most direct way. Thus, the surveys of informal businessowners may include questions on the costs and benefits of formality and informality, the degree to which they are affected by control mechanisms and corruption – factors that might incentivize the transition to formality, the perception of the simplicity or complexity of the procedures for this transition, the availability of programmes and institutions that support it and the information that businessowners have on relevant issues, such as the legal regulations on formality or informality.

The inclusion of owners of recently formalized businesses permits coverage of the experiences of the transition to formality and with formality (see Coolidge and Ilic 2009). The questions raised in this context may refer to the reasons that led the owner to opt for the transition to formality, the relevance, effectivity and efficiency of the policies and institutions that fostered the transition to formality and the impact of this transition on the success of the business.

Qualitative surveys that cover informal employees may include questions on the advantages and disadvantages of informality and formality, their interest in formalization, their familiarity with labour laws, employment promotion policies and programmes and the participation in these programmes, as well as the economic and personal obstacles they perceive regarding the transition to formality.

The combination of the answers to qualitative questions with quantitative data helps to understand the diversity of the informal economy in terms of, for example, the size of businesses (value added, number of workers), industries, the premises a business operates on and the original motive for the operation of the business, the duration of its existence, as well as personal characteristics of the businessowner, such as education, age, gender or migration status.

The application of econometric techniques permits analysis of factors that incentivize and hamper businesses´ transition to formality. For example, Nguyen, Verreynne and Steen (2014) analysed the role

of three drivers in Vietnamese firms´ decision to formalize during a period of six years: the desire to grow and innovate; the desire to obtain access to government financial support; and the desire to avoid paying bribes. Tran and La (2020) analysed the role of government regulation and corruption on this same decision. In both cases, a set of control variables (firm performance, industry, location and the characteristics of the owner and the workforce) were taken into account. Tran and La also included a "switching variable" that distinguished firms that had formalized after the first survey round from the ones that had not.

The econometric analysis of consecutive surveys also permits inquiry on the impact of formalization on business performance.

Regression analysis has been applied to better understand whether and to what degree GDP per capita, economic and labour productivity growth, employment protection regulation, business flexibility, the enforcement of and compliance with regulations, the tax burden, labour market participation, the unemployment rate, the union affiliation rate and legislative measures that promote the transition to formality influence the size of informal employment (see Salazar-Xirinachs and Chacaltana 2018 and Loayza and Rigolini 2006).

Methods that decompose changes in the rate of job formality on an aggregate level aim to understand the relative weight of economic factors (basically, economic growth and the structural change related to it) represented by the between-effect and the result of other factors, such as policy interventions to foster the transition to formality, represented by the within-effect.

▶ Pros and cons

Pros

- Quantitative surveys and, especially, the econometric use of the data they produce, permit detailed analysis of the characteristics and, under certain circumstances, the dynamics of the informal economy.
- If adequate data are available, the heterogeneity of informality can be integrated into regression models, and the different characteristics and correlations can be considered for policy advice.
- The 1-2-3 surveys build a sample at the household level and can be used to identify so-called "invisible" informal economic units, especially in the informal sector's lower tier.
- Satellite accounts cover the informal sector and informal employment outside the informal sector to overcome unrealistic assumptions on fixed proportions of the informal economy that characterize standard national account macrodata.
- Qualitative surveys can contribute to the understanding of the characteristics of the informal economy and its heterogeneity as well as the role of public policies.
- Decomposition methods can distinguish the effect of structural change and patterns of economic growth on the rate of job formality or informality.

Cons

- Samples for quantitative enterprise surveys cannot cover a complete universe because not all informal economic units are included in databases on enterprises, which limits their representativity. Although the 1-2-3 surveys can confront this challenge by identifying informal businesses not included in any enterprise directory, they may introduce another bias due to the use of the population census for the construction of the sampling universe.
- Satellite accounts and regression models can indicate changes of the relevant variables but cannot present causalities (although certain methods may strengthen arguments in favour of causal relations).
- Qualitative surveys occasionally exclude economic units in the lower tier of the informal economy intentionally because their transition to formality is seen as unrealistic, which can lead to errors in the analysis of the dynamics of the informal economy as a whole.
- The results of qualitative surveys may not be trustworthy because respondents who are aware of their illegal status may provide biased answers to certain questions.
- The within-effects in a decomposition of changes in the rate of formality and informality represent a "black box" because the underlying factors are unknown. There could be composition effects within industries (between subsectors or firms of different size). Structural change is represented by industries´ participation in total employment, which might be influenced by an unequal evolution of informality between industries, without necessarily representing changes in the productive structure.

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